DESIGN AND ANALYSIS OF COMPACT MICROSTRIP RESONATOR-BASE BANDPASS FILTER FOR WIRELESS COMMUNICATION APPLICATIONS

A Project report submitted in partial fulfillment of the requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS & COMMUNICATION ENGINEERING

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CERTIFICATE

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We certify that the work contained in this report is original and has been done by us under the guidance of my supervisor. The work has not been submitted to any other Institute for any degree or diploma. We have followed the guidelines provided by the Institute in preparing the report. We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute. Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.

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ACKNOWLEDGEMENT

We are tremendously obliged to our guide **Dr. M. NAVEEN KUMAR,** Assistant Professor, Department of Electronics and Communication Engineering, **POTTI SRIRAMULU CHALAVADI MALLIKARJUNA RAO COLLEGE OF ENGINEERING & TECHNOLOGY** for his patience guidance, in valuable advice and selfless help rendered all through the progress of our work.

We sincerely thank the Hon'ble Secretary & correspondent **Dr. C. A. CHITTA AMAR SUDHEER**, for providing us with the good facilities in our college for carrying out the project successfully, and to our Principal **Dr. S. SARAVANA KUMAR** for their kind encouragement and their genial cooperation.

We are extremely grateful to our Head of the Department, **Dr. A. Bala Krishna**, for his constant encouragement and invaluable support throughout our thesis work. We also extend our heartfelt **thanks to all the teaching and non-teaching staff** of the ECE Department, and to the entire staff of Potti Sriramulu Chalavadi Mallikarjuna Rao College of Engineering and Technology, for their cooperation and assistance.

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PEO	Objective
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PEO 2 (Core Competence):	To motivate graduates to Analyze, Design, Develop, Optimize and Implement electronic systems with competent spirit.
PEO3 (Breadth Knowledge):	To enable graduates with sufficient breadth in electronics and its related fields to solve general engineering problems in an eco-friendly environment.
PEO 4 (Soft Skills):	To make graduates with a professional outlook where it can communicate effectively and interact responsibly with colleagues, clients, employers and society.
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IV Year II Semester Project Course Outcomes

CO's	Course Outcomes	BTL
CO1	Identify the complex engineering problems relevant to the society and industry	Apply
CO2	Apply modern technologies, tools, and systems in the field of Electronics & Communication Engineering to analyze the identified problem	Apply
CO3	Design and implement a viable solution to the problem	Create
CO4	Apply Communication & Presentation skills	Apply
CO5	Develop the team work, leadership skills with professional and ethical values.	Understand
CO6	Make the use of report writing skills	Understand

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ABSTRACT

This paper presents the design and implementation of a Band-pass filter (BPF) for ISM band wireless communication applications. This Bandpass filter uses a modified resonator structure with optimized coupling and compact microstrip layout to improve the filter performance. Conventional band-pass filters implemented on FR-4 substrates typically exhibit higher dielectric losses, limited selectivity, and restricted bandwidth. To address these limitations, the proposed design employs a Rogers RT/Duroid 5880 substrate, known for its low dielectric loss and suitability for high- frequency microwave circuits. The filter is modeled and analyzed using ANSYS HFSS simulation software. The use of an optimized substrate significantly enhances signal transmission characteristics, reduces power dissipation, and improves overall filter efficiency. The proposed design is well-suited for RF front-end modules, wireless transceivers, and IoT-based communication systems operating in the ISM band. The results demonstrate that appropriate substrate selection plays a crucial role in achieving high-performance and reliable microwave filter designs.

Keywords: Band Pass Filter(BPF),U-Shape Coupling, ISM Band, Rogers RT/Duroid 5880, Microwave filter , RF Front-end ,Wireless Communication, HFSS

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LIST OF ABBREVIATIONS

Abbreviation	-	**Description**
IOT	-	Internet of Things
BW	-	Bandwidth
WBAN	-	Wireless Body Area Networks
HFSS	-	High-Frequency Structure Simulator
LPF	-	Low Pass Filter
HPF	-	High Pass Filter
RF	-	Radio Frequency
GSM	-	Global System for Mobile Communications
LTE	-	Long-Term Evolution
2G	-	Second Generation
3G	-	Third Generation
4G	-	Fourth Generation
5G	-	Fifth Generation
VSWR	-	Voltage Standing Wave Ratio
RL	-	Return Loss
GP	-	Ground Plane
EM	-	Electromagnetic
BSF	-	Band Stop Filter
BPF		Band Pass Filter

CHAPTER 1

INTRODUCTION

1.1 Introduction

In modern wireless communication systems, filters play a crucial role in ensuring efficient and reliable signal transmission. With the rapid advancement of technologies such as wireless networks, satellite communication, and Internet of Things (IoT) systems, there is a growing demand for compact, high-performance, and low-loss filtering components. Filters are essential for selecting desired frequency components while rejecting unwanted signals, noise, and interference, thereby improving overall system performance.

A filter is an electronic circuit designed to pass specific frequency ranges and attenuate others. In radio frequency (RF) and microwave communication systems, filters are widely used in transmitters and receivers to maintain signal integrity and prevent interference between different communication channels. Among various types of filters, the band-pass filter (BPF) is one of the most important, as it allows only a specific range of frequencies to pass while blocking frequencies outside this range.

This thesis focuses on the design and analysis of a compact microstrip resonator-based band-pass filter for wireless communication applications. Microstrip filters are widely preferred due to their advantages such as low profile, lightweight structure, ease of fabrication, and compatibility with integrated circuits. However, conventional microstrip filters implemented on FR-4 substrates often suffer from higher dielectric losses, limited bandwidth, and poor selectivity at high frequencies.

To overcome these limitations, the proposed design utilizes a Rogers RT/Duroid 5880 substrate, which offers low dielectric loss, high efficiency, and better performance in microwave frequency applications. The filter design incorporates a modified resonator structure with optimized coupling techniques to enhance impedance matching, reduce

insertion loss, and improve bandwidth characteristics.

The design and analysis are carried out using ANSYS HFSS (High-Frequency Structure Simulator), which provides accurate electromagnetic simulation and optimization of RF components. The performance of the filter is evaluated using key parameters such as return loss (S11), insertion loss (S21), Voltage Standing Wave Ratio (VSWR), bandwidth, center frequency, and quality factor.

The proposed filter operates around the microwave frequency range and is suitable for S applications in wireless communication systems, RF front-end modules, and IoT-based devices. The results demonstrate that proper design and substrate selection significantly improve filter performance, making it suitable for modern high-frequency communication systems.

1.2 Motivation of Thesis

The rapid evolution of wireless communication technologies has significantly increased the demand for high-performance, compact, and efficient RF components. Modern communication systems such as Wi-Fi, 5G, satellite communication, and Internet of Things (IoT) applications require reliable signal transmission with minimal losses and interference. Among these components, filters play a vital role in ensuring proper frequency selection and maintaining signal integrity.

In practical wireless environments, multiple signals operate simultaneously over different frequency bands. This often leads to interference, noise, and signal distortion, which degrade system performance. Therefore, there is a strong need for efficient filtering mechanisms that can selectively pass desired frequency bands while rejecting unwanted signals.

Conventional band-pass filters designed using standard substrates such as FR-4 suffer from several limitations, especially at higher frequencies. These limitations include high dielectric losses, poor quality factor, limited bandwidth, and inadequate impedance matching. As a result, the overall efficiency and reliability of the communication system are reduced. Moreover, many traditional filter designs are bulky in size and lack compactness, making them unsuitable for modern miniaturized electronic devices.

Another major challenge in filter design is achieving a balance between key performance parameters such as insertion loss, return loss, bandwidth, and selectivity. Improving one parameter often leads to degradation in another, making the design process more complex. Additionally, achieving stable performance over the desired frequency range requires careful optimization of the filter structure and materials.

To address these challenges, there is a need to develop advanced filter designs that offer improved performance while maintaining compact size and ease of fabrication. The use of low-loss substrates such as Rogers RT/Duroid 5880 provides better dielectric properties, reducing signal attenuation and enhancing overall efficiency. Furthermore, incorporating modified resonator structures and optimized coupling techniques can significantly improve filter characteristics such as bandwidth, impedance matching, and selectivity.

The motivation behind this thesis is to design and analyze a compact microstrip resonator-based band-pass filter that overcomes the limitations of conventional designs. The proposed

approach focuses on enhancing performance by selecting appropriate materials, optimizing geometrical parameters, and utilizing advanced simulation tools like ANSYS HFSS.
This work aims to contribute to the development of efficient and reliable filter designs suitable for modern wireless communication systems. By improving key performance metrics and reducing losses, the proposed filter can be effectively used in RF front-end modules, wireless transceivers, and IoT-based applications, thereby supporting the growing demands of next-generation communication technologies.

1.3 Objective of Thesis

The primary objective of this thesis is to design and analyze a compact and efficient microstrip resonator-based band-pass filter for wireless communication applications operating in the microwave frequency range. The proposed work aims to overcome the limitations of conventional filter designs by improving important performance parameters such as return loss, insertion loss, bandwidth, and impedance matching. Special emphasis is given to achieving low signal reflection and high transmission efficiency, ensuring reliable communication performance.

Another key objective of this work is to achieve proper impedance matching and maintain a low Voltage Standing Wave Ratio (VSWR), which indicates efficient power transfer and minimal energy loss in the system. In order to enhance high-frequency performance and reduce dielectric losses, a low-loss substrate, namely Rogers RT/Duroid 5880, is utilized instead of conventional FR-4 material. The design also focuses on incorporating a modified resonator structure with optimized coupling techniques to improve selectivity, compactness, and overall filter efficiency.

Furthermore, the thesis aims to model and simulate the proposed filter using ANSYS HFSS, which enables accurate electromagnetic analysis and optimization of the design. Various performance parameters such as bandwidth, center frequency, quality factor, field distributions, and Smith chart characteristics are analyzed to evaluate the effectiveness of

the proposed filter. The work also includes a comparison with existing filter designs to demonstrate improvements in performance.

1.4 Organization of Thesis

The thesis is systematically structured into seven chapters, each addressing critical aspects of the design and analysis of a compact microstrip resonator-based band-pass filter for wireless communication applications.

Chapter1:Introduction

This chapter provides a comprehensive overview of band-pass filters, highlighting their importance in modern wireless communication systems. It discusses the need for efficient filtering, motivation behind the proposed work, and the objectives of the thesis.

Chapter 2: Literature Survey

This chapter presents a detailed review of existing research related to band-pass filter design techniques, resonator-based structures, and performance enhancement methods. It also includes a comparative analysis of key parameters such as return loss (S11), insertion loss (S21), bandwidth, and selectivity.

Chapter 3: Introduction to Filters

This chapter introduces the basic concepts and characteristics of filters, including types of filters, frequency response, bandwidth, and quality factor. Special focus is given to band-pass filters, their working principles, structures, advantages, and applications in RF and microwave systems.

Chapter 4: HFSS Software and Design Methodology

This chapter focuses on the design and simulation of the proposed band-pass filter using ANSYS HFSS (High-Frequency Structure Simulator). It includes details of substrate selection, material properties, resonator design, boundary conditions, and simulation setup.

Chapter 5: Implementation of Proposed Band-Pass Filter

This chapter describes the step-by-step implementation of the proposed microstrip

resonator-based band-pass filter. It covers the design procedure, resonator structure, coupling mechanism, and optimization techniques used to achieve improved performance.

Chapter 6: Results and Discussion

This chapter presents the simulation results and performance evaluation of the proposed filter. Key parameters such as return loss (S11), insertion loss (S21), VSWR, bandwidth, center frequency, and field distributions are analyzed and discussed in detail.

Chapter 7: Conclusion and Future Scope

This chapter summarizes the overall work and highlights the key contributions of the proposed design. It also suggests future improvements such as miniaturization, multi-band filter design, and integration with advanced wireless communication systems.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The rapid advancement of wireless communication technologies has significantly increased the demand for efficient, compact, and high-performance radio frequency (RF) and microwave components. Among these, filters play a crucial role in ensuring reliable communication by allowing desired frequency signals to pass while rejecting unwanted signals, noise, and interference. Band-pass filters (BPFs) are widely used in applications such as Wi-Fi, Bluetooth, satellite communication, radar systems, and Internet of Things (IoT) devices, where precise frequency selection is essential for proper system operation. A band-pass filter is designed to allow a specific range of frequencies, known as the passband, while attenuating frequencies outside this range. The performance of a BPF is determined by parameters such as center frequency, bandwidth, insertion loss, return loss, selectivity, and quality factor. Achieving optimal performance across all these parameters is a key challenge in filter design, especially in modern communication systems where multiple frequency bands operate simultaneously, leading to interference and signal degradation.

Microstrip band-pass filters have become popular due to their compact size, low profile, ease of fabrication, and compatibility with printed circuit board (PCB) technology. However, their performance is influenced by several factors, including substrate material, resonator structure, and coupling mechanism. Conventional substrates such as FR-4 are widely used due to their low cost, but they exhibit higher dielectric losses at microwave frequencies, which reduces efficiency. To overcome this limitation, low-loss substrates such as Rogers RT/Duroid 5880 are preferred, as they provide better performance and reduced signal attenuation.

The choice of resonator structure also plays a significant role in determining filter performance. Various resonator configurations such as ring resonators, hairpin structures, and coupled-line resonators have been widely studied to improve selectivity, bandwidth, and compactness. Additionally, advanced techniques such as defected ground structures (DGS) and electromagnetic bandgap (EBG) structures are used to enhance impedance matching and frequency response by modifying the electromagnetic characteristics of the filter.

With the increasing demand for compact and high-performance wireless devices, there is

a growing need for filters that offer improved bandwidth, low insertion loss, and better selectivity while maintaining a small size. Advanced simulation tools such as ANSYS HFSS enable accurate modeling and optimization of filter designs, allowing designers to achieve desired performance before fabrication.

Despite significant advancements, challenges such as achieving low loss, high selectivity, and compact size simultaneously still exist. Therefore, this work focuses on developing an efficient microstrip resonator-based band-pass filter that addresses these challenges and meets the requirements of modern wireless communication systems.

2.2 Literature Survey

Varshney et al. presented a novel low-loss impedance transformer-less fish-tail-shaped microstrip-to-waveguide (MS-to-WG) transition designed for high-frequency applications across K-, Ka-, Q-, and U-bands, addressing one of the critical challenges in microwave and millimeter-wave system integration. In conventional MS-to-WG transitions, impedance matching is typically achieved using stepped impedance transformers or additional matching circuits, which often introduce increased insertion loss, structural complexity, and fabrication challenges. To overcome these limitations, the authors proposed a transformer-less transition design based on a fish-tail-shaped microstrip geometry that ensures smooth electromagnetic field transformation between the planar microstrip line and the waveguide structure. The fish-tail configuration was carefully optimized to gradually distribute the electromagnetic fields, thereby minimizing abrupt discontinuities that usually cause reflection losses and impedance mismatch. This innovative design approach significantly reduces insertion loss while maintaining a wide operational bandwidth, which is essential for modern high-frequency communication systems. The authors conducted a detailed parametric analysis to evaluate the impact of geometrical parameters such as taper length, width variation, and substrate properties on the overall performance of the transition. Simulation and experimental results demonstrated excellent agreement, showing low return loss and high transmission efficiency across the targeted frequency bands. The proposed structure also exhibited superior performance compared to traditional transition designs, particularly in terms of reduced reflection coefficient and improved bandwidth. Furthermore, the compact and simple structure makes it highly suitable for integration into advanced RF and microwave systems, including satellite communication, radar systems, and next-generation wireless networks. The elimination of impedance transformers not only simplifies the design but also enhances reliability and reduces fabrication cost. This work highlights the importance of innovative geometrical design in achieving high-performance microwave components and provides a valuable contribution to the development of efficient interconnects in high-frequency applications. [1]

Yu et al. proposed an advanced feedback-type RF oscillator integrated with a modified magnetic-LC resonator for microwave sensing applications, specifically targeting the characterization of liquid samples with high precision. The primary motivation behind this

work was to improve the sensitivity and stability of microwave sensing systems, which are often limited by noise and frequency instability in conventional designs. The authors utilized a magnetic-LC resonator structure due to its strong electromagnetic field confinement and high quality factor, which enhances the interaction between the electromagnetic waves and the material under test. By incorporating a feedback mechanism into the oscillator design, the system achieved stable oscillations and improved frequency tracking capability. The sensing principle is based on the shift in resonant frequency caused by variations in the dielectric properties of the liquid sample placed near the resonator. Even minor changes in permittivity resulted in measurable frequency deviations, demonstrating high sensitivity and accuracy. The authors also performed experimental validation, confirming that the proposed system exhibits excellent repeatability and reliability under varying conditions. Furthermore, the compact design allows easy integration into portable sensing devices, making it suitable for real-time applications such as biomedical diagnostics, chemical analysis, and environmental monitoring. The study highlights the potential of combining resonator-based structures with feedback oscillator techniques to develop high-performance microwave sensors, thereby extending the application of RF systems beyond communication into interdisciplinary domains. [2]

Qu et al. introduced a photonic delay reservoir computing system based on ring resonator structures for reconfigurable microwave waveform generation, representing a significant advancement in the integration of photonic and microwave technologies. The proposed system leverages the inherent delay and nonlinear characteristics of optical ring resonators to implement reservoir computing, which is a form of neural network capable of processing time-dependent signals. Unlike traditional electronic signal processing systems, the photonic approach offers advantages such as high speed, large bandwidth, and immunity to electromagnetic interference. The authors demonstrated that by carefully designing the ring resonator parameters, it is possible to generate complex microwave waveforms with high precision and flexibility. The system also exhibited reconfigurability, allowing it to adapt to different signal processing requirements without significant hardware modifications. Experimental results showed improved performance in terms of signal quality, stability, and processing speed compared to conventional approaches. Additionally, the compact nature of photonic integrated circuits makes the proposed design suitable for next-generation communication systems where size, speed, and efficiency are critical. This work highlights the growing trend of hybrid photonic-microwave systems and provides valuable insights into the future of high-speed signal processing and

communication technologies. [3]

Rabbani et al. proposed a modified circular ring resonator-based metamaterial absorber incorporating a dumbbell-shaped structure, designed to achieve high absorption efficiency across S, X, and Ku frequency bands. The study focused on addressing the limitations of conventional absorbers, which often suffer from narrow bandwidth and limited absorption efficiency. By integrating a dumbbell-shaped structure into the ring resonator, the authors enhanced the electromagnetic response, leading to improved absorption characteristics. The design operates based on the principle of resonance, where both electric and magnetic fields are strongly confined within the structure, resulting in near-perfect absorption at specific frequencies. The authors conducted a detailed analysis of the geometrical parameters and their impact on absorption performance, demonstrating that careful optimization can significantly enhance bandwidth and efficiency. The proposed absorber also exhibited polarization insensitivity and angular stability, making it suitable for practical applications such as radar cross-section reduction, electromagnetic shielding, and sensing systems. Experimental validation confirmed the effectiveness of the design, showing good agreement with simulation results. This work demonstrates the potential of combining metamaterial concepts with resonator structures to develop advanced electromagnetic absorbers for modern wireless and defense applications. [4]

Dutta et al. presented a three-dimensional frequency selective surface (FSS) design featuring multiple transmission zeros and a pseudo-elliptic response, aimed at improving selectivity and frequency response in microwave filtering applications. Traditional FSS designs often suffer from limited selectivity and poor out-of-band rejection, which restrict their performance in high-frequency systems. To address these issues, the authors introduced multiple transmission zeros into the design, resulting in sharp roll-off characteristics and enhanced stopband performance. The pseudo-elliptic response further improved the filter's ability to discriminate between desired and undesired frequencies. The three-dimensional structure provided additional degrees of freedom in design, allowing better control over electromagnetic wave propagation. The authors also performed parametric studies to analyze the influence of structural parameters on performance, providing valuable design guidelines. The proposed FSS demonstrated improved bandwidth, reduced interference, and enhanced overall performance compared to conventional designs. This work highlights the importance of advanced structural configurations in achieving high-performance filtering characteristics and contributes to the development of next-generation frequency selective devices. [5]

Cheng introduced the concept of dual-mode ring resonators, which became a significant breakthrough in the design of compact band-pass filters. The study focused on utilizing degenerate modes within a single resonator structure to achieve dual-mode operation, thereby reducing the number of resonators required in the filter design. This approach not only minimized the overall size of the filter but also improved its performance by introducing transmission zeros, which enhance selectivity and provide better out-of-band rejection. The author demonstrated that by carefully perturbing the symmetry of the ring resonator, it is possible to split the degenerate modes and control the frequency response effectively. This method enabled the design of filters with sharper roll-off characteristics and improved stopband attenuation. Additionally, the proposed design provided flexibility in tuning the filter parameters, making it suitable for a wide range of applications in microwave communication systems. The work laid the foundation for many subsequent studies on dual-mode resonator-based filters and remains a key reference in modern filter design methodologies. [6]

Ren et al. developed a high-order high-temperature superconducting (HTS) dual-band differential band-pass filter using a stub-loaded twin-ring resonator structure. The proposed design aimed to achieve high performance in terms of low insertion loss, high selectivity, and dual-band operation. By incorporating stub-loaded resonators, the authors were able to control the passband frequencies independently, allowing precise tuning of the filter response. The differential configuration provided improved noise immunity and reduced electromagnetic interference, which is essential for high-frequency communication systems. The use of HTS materials further enhanced the performance by significantly reducing conductor losses, resulting in superior quality factor and efficiency. Experimental results demonstrated excellent agreement with simulations, confirming the effectiveness of the design. This work highlights the importance of combining advanced materials with innovative resonator structures to achieve high-performance filters for next-generation communication systems. [7]

Jafari and Malekshahi proposed a low-cost microwave microfluidic sensor based on a planar ring resonator, focusing on the detection and characterization of liquid samples. The design utilized the sensitivity of the resonant frequency to changes in dielectric properties of the material under test. By integrating a microfluidic channel with the resonator structure, the authors were able to achieve real-time sensing with high accuracy. The planar configuration allowed easy fabrication and integration with existing microwave circuits. The study demonstrated that small variations in the permittivity of the liquid resulted in

measurable frequency shifts, enabling precise characterization. The proposed sensor also showed good repeatability and stability, making it suitable for practical applications in biomedical and chemical analysis. This work emphasizes the versatility of ring resonator structures, extending their application beyond communication systems into sensing technologies. [8]

Salleh et al. presented a quarter-wavelength side-coupled ring resonator design for band-pass filter applications, focusing on achieving compact size and improved coupling efficiency. The proposed structure utilized side coupling between the feed lines and the ring resonator to enhance energy transfer and reduce losses. The quarter-wavelength configuration allowed significant size reduction compared to traditional half-wavelength resonators, making it suitable for compact microwave circuits. The authors also analyzed the impact of coupling coefficients and resonator dimensions on filter performance, demonstrating that proper optimization can lead to improved bandwidth and selectivity. Experimental results confirmed the effectiveness of the design, showing low insertion loss and good return loss characteristics. This work contributed significantly to the development of compact and efficient resonator-based filters. [9]

Cheng further extended his work on dual-mode ring resonators by exploring different configurations for generating transmission zeros and improving filter selectivity. The study emphasized the importance of mode coupling and symmetry perturbation in controlling the frequency response of the filter. By adjusting the resonator geometry, the author demonstrated that it is possible to achieve multiple transmission zeros, leading to enhanced stopband performance. This approach provided a systematic method for designing high-selectivity filters with compact size. The findings of this study have been widely adopted in modern filter design and have influenced numerous research works in the field of microwave engineering. [10]

Matsuo et al. proposed dual-mode stepped impedance ring resonators for band-pass filter applications, focusing on improving harmonic suppression and stopband performance. The stepped impedance structure allowed better control over the distribution of electric and magnetic fields within the resonator, enabling suppression of unwanted harmonic frequencies. The dual-mode operation further enhanced the efficiency of the design by reducing the number of required resonators. The authors demonstrated that the proposed structure achieved improved selectivity and reduced insertion loss compared to conventional designs. The study also provided detailed analysis and design guidelines, making it a valuable reference for researchers working on advanced filter design

techniques. [11]

Hsieh and Chang introduced dual-mode quasi-elliptic band-pass filters using ring resonators, aiming to achieve sharp roll-off characteristics and improved selectivity. The proposed design incorporated transmission zeros near the passband edges, resulting in enhanced stopband attenuation. The quasi-elliptic response allowed better control over the frequency response, making the filter suitable for applications requiring high selectivity. The authors also discussed the impact of coupling mechanisms and resonator parameters on filter performance, providing a comprehensive understanding of the design process. This work has been widely cited in the literature and remains a significant contribution to the field of microwave filter design. [12]

Gorur proposed a dual-mode band-pass filter design with Chebyshev and elliptic responses, offering flexibility in shaping the frequency response. The study demonstrated that by adjusting the resonator parameters, it is possible to achieve different types of filter responses depending on application requirements. The proposed design provided improved selectivity and reduced insertion loss, making it suitable for high-performance communication systems. The ability to switch between different response types adds versatility to the filter design, making it adaptable to various applications. [13]

Chang and Hsieh provided a comprehensive analysis of microwave ring circuits and related structures, covering both theoretical and practical aspects of resonator-based design. Their work included detailed discussions on ring resonator theory, coupling mechanisms, and filter synthesis techniques. The book serves as a fundamental reference for understanding the principles of microwave filter design and has been widely used by researchers and engineers in the field. It provides valuable insights into the design and optimization of high-performance filters using ring resonators. [14]

He et al. demonstrated a chip-scale photonic microwave oscillator, highlighting the integration of photonic and microwave technologies for high-frequency signal generation. The proposed system achieved improved stability and reduced phase noise compared to conventional electronic oscillators. The compact size and high performance make it suitable for advanced communication systems and signal processing applications. [15]

Kudelin et al. developed a photonic chip-based microwave oscillator, further advancing the integration of photonic technologies in RF systems. The design demonstrated high efficiency and stability, making it suitable for next-generation communication systems. This work highlights the potential of photonic approaches in improving microwave system performance. [16]

Yu et al. proposed a magnetic-LC resonator-based microwave sensing system designed to achieve high sensitivity and precision in the characterization of liquid samples. The system operates on the principle that variations in the dielectric properties of the material under test directly influence the resonant frequency of the LC resonator. By integrating a feedback-type RF oscillator, the authors ensured stable oscillations and enhanced detection capability, which is critical for accurate sensing. The magnetic-LC resonator structure provides strong electromagnetic field confinement, thereby increasing interaction with the sample and improving sensitivity. The study also investigated the influence of structural parameters such as inductance, capacitance, and resonator geometry on overall performance. Experimental results confirmed that even slight changes in permittivity could be detected with high accuracy, demonstrating excellent repeatability and reliability. Furthermore, the compact size and simple configuration make the system suitable for portable sensing devices. This work highlights the growing importance of microwave sensing technologies in interdisciplinary applications such as biomedical diagnostics, chemical analysis, and environmental monitoring. [17]

Qu et al. explored the implementation of photonic delay reservoir computing using ring resonator structures for advanced microwave waveform generation and signal processing. The proposed system leverages the inherent delay properties and nonlinear dynamics of optical ring resonators to process time-dependent signals efficiently. Unlike conventional electronic processing systems, the photonic approach offers significantly higher bandwidth, faster processing speed, and immunity to electromagnetic interference. The authors demonstrated that the system could generate complex microwave waveforms with high precision by tuning the parameters of the photonic reservoir. Additionally, the reconfigurable nature of the design allows it to adapt to different signal processing tasks without requiring major hardware modifications. The integration of photonic and microwave technologies represents a significant advancement in RF system design, enabling next-generation communication systems to achieve higher performance and flexibility. Experimental validation showed improved accuracy and stability compared to traditional methods, emphasizing the potential of photonic computing in microwave engineering. [18]

Rabbani et al. developed a circular ring resonator-based metamaterial absorber capable of achieving high absorption efficiency across multiple frequency bands. The proposed design

utilizes the unique electromagnetic properties of metamaterials to trap and dissipate incident electromagnetic energy effectively. By carefully designing the geometry of the ring resonator, the authors achieved strong electric and magnetic resonance coupling, resulting in near-perfect absorption at specific frequencies. The study also examined the impact of structural parameters such as resonator dimensions and substrate properties on absorption performance. The absorber demonstrated polarization insensitivity and maintained stable performance over a wide range of incident angles, making it suitable for practical applications such as radar cross-section reduction and electromagnetic interference shielding. The compact and multi-band nature of the design further enhances its applicability in modern wireless systems. This work underscores the importance of combining resonator structures with metamaterial concepts to develop advanced electromagnetic components. [19]

Mosbah et al. proposed a microwave liquid sensor based on split-ring resonators, focusing on achieving high sensitivity and accurate detection of liquid properties. The sensing mechanism is based on the shift in resonant frequency caused by changes in the dielectric constant of the liquid under test. The split-ring resonator structure provides strong field confinement, which enhances interaction with the sample and improves sensitivity. The authors conducted a detailed analysis of the influence of resonator geometry and substrate material on sensor performance, demonstrating that proper optimization can significantly improve accuracy. Experimental results showed that the sensor could distinguish between different liquid samples with high precision and repeatability. The planar design allows easy fabrication and integration with existing microwave systems, making it suitable for practical applications. This work contributes to the advancement of microwave sensing technologies and highlights the effectiveness of split-ring resonators in real-world applications. [20]

Jafari and Malekshahi further investigated planar ring resonator-based microwave sensors with a focus on low-cost implementation and high performance. The proposed design achieved improved sensitivity by optimizing the resonator structure and enhancing electromagnetic interaction with the material under test. The authors demonstrated that the sensor could accurately detect variations in dielectric properties, making it suitable for applications in material characterization and chemical analysis. The study also emphasized the importance of fabrication simplicity, enabling large-scale production without compromising performance. The results showed high repeatability and reliability,

confirming the practicality of the design. This work highlights the continued relevance of resonator-based sensors in modern sensing technologies and their potential for widespread adoption. [21]

Khobzei et al. introduced an adjustable band-pass filter using ring resonators, focusing on achieving tunable frequency response for modern communication systems. The proposed design allows dynamic control of the passband frequency by modifying resonator parameters, enabling flexible operation across different frequency bands. The authors demonstrated that the filter maintained good performance characteristics, including low insertion loss and high selectivity, even under varying tuning conditions. The ability to adjust frequency response in real-time makes the design highly suitable for applications such as cognitive radio and adaptive communication systems. This work emphasizes the importance of reconfigurable filter designs in meeting the evolving requirements of wireless communication technologies. [22]

Li et al. proposed a miniaturized frequency selective rasorber designed for 5G applications, combining both absorption and filtering functionalities in a single compact structure. The design achieved effective suppression of unwanted signals while maintaining a small form factor, which is essential for modern wireless devices. The authors utilized advanced materials and structural optimization techniques to enhance electromagnetic performance. The proposed rasorber demonstrated improved bandwidth, reduced interference, and high efficiency, making it suitable for next-generation communication systems. This work highlights the trend toward multifunctional RF components that integrate multiple capabilities into a single device. [23]

Liu et al. developed a wideband frequency-selective structure using split-ring resonators, focusing on improving bandwidth and selectivity. The proposed design utilized the unique properties of split-ring resonators to achieve wideband operation while maintaining strong filtering characteristics. The authors conducted a detailed parametric study to analyze the influence of structural parameters on performance, providing valuable design insights. The results demonstrated improved bandwidth and reduced signal distortion, making the design suitable for high-speed communication systems. This work contributes to the advancement of wideband filter technologies. [24]

Gultekin and Yerlikaya proposed a compact antenna integrated with a ring-shaped

frequency selective surface (FSS), aiming to enhance antenna performance while simultaneously incorporating filtering characteristics within the same structure. The primary objective of their work was to reduce system complexity by eliminating the need for separate filtering components, thereby achieving a more compact and efficient RF front-end design. The ring-shaped FSS was strategically placed to control electromagnetic wave propagation, enabling selective frequency transmission and rejection. The authors conducted a detailed analysis of the antenna parameters, including gain, bandwidth, radiation pattern, and impedance matching, demonstrating significant improvements compared to conventional antenna designs. The integration of FSS also contributed to reduced mutual coupling and enhanced isolation, which are critical in multi-element systems. Furthermore, the compact size and lightweight structure make the design suitable for modern wireless communication systems, particularly in portable and embedded applications. The study highlights the growing trend of integrating multiple functionalities into a single device to improve overall system efficiency and performance. This work provides valuable insights into the design of compact, high-performance antenna systems using frequency selective surfaces. [25]

Zeain et al. presented an advanced frequency selective surface (FSS)-based compact multiple-input multiple-output (MIMO) antenna designed specifically for 5G communication systems. The main focus of their research was to improve isolation between antenna elements while maintaining high efficiency and wide bandwidth, which are essential requirements for MIMO systems. The authors incorporated FSS structures to suppress unwanted coupling effects and enhance overall antenna performance. The design demonstrated improved gain, reduced interference, and better signal quality, making it suitable for high-speed data transmission applications. The study also explored the impact of FSS geometry and placement on antenna characteristics, providing a comprehensive understanding of the design process. Experimental results showed excellent agreement with simulations, confirming the effectiveness of the proposed approach. The compact and efficient design makes it highly suitable for integration into modern wireless devices, where space constraints and performance requirements are critical. This work emphasizes the importance of combining filtering and antenna technologies to achieve

optimized RF system performance. [26]

Hossain et al. developed advanced miniaturized artificial materials based on split-ring resonator (SRR) structures, focusing on enhancing electromagnetic characteristics such as effective permittivity and permeability for microwave and RF applications. The study emphasized the importance of metamaterial-inspired designs in achieving compactness while maintaining superior electromagnetic performance. By utilizing SRR structures, the authors were able to engineer artificial materials that exhibit negative or enhanced electromagnetic parameters, which are not readily available in natural materials. These engineered properties allow better control over electromagnetic wave propagation, enabling improved confinement and manipulation of signals in microwave circuits. The authors performed an extensive parametric analysis to study the influence of geometrical dimensions, substrate properties, and resonator configurations on the overall performance of the material. The results demonstrated that the proposed artificial material could significantly enhance device miniaturization while maintaining high efficiency. Additionally, the study highlighted potential applications in filters, antennas, absorbers, and sensing systems, where improved electromagnetic properties are essential. The compact size and improved functionality make these materials highly suitable for integration into modern wireless communication systems, particularly in applications requiring lightweight and space-efficient designs. This work contributes significantly to the field of metamaterials and provides a strong foundation for the development of next-generation microwave components with enhanced performance and reduced size. [27]

Zhang et al. proposed a tunable microwave photonic band-pass filter that integrates optical and microwave technologies to achieve dynamic control over frequency response, representing a major advancement in reconfigurable RF systems. The proposed design utilizes photonic components such as optical modulators and delay lines to manipulate microwave signals in the optical domain, thereby enabling high-speed processing and wide bandwidth operation. One of the key advantages of this approach is the ability to achieve tunability without physically altering the circuit structure, which significantly enhances flexibility and adaptability. The authors demonstrated that by adjusting optical parameters such as wavelength and phase,

the center frequency and bandwidth of the filter can be dynamically controlled. This capability is particularly useful in modern communication systems that require real-time frequency reconfiguration, such as cognitive radio and adaptive networks. The study also highlighted the benefits of photonic systems, including low insertion loss, high immunity to electromagnetic interference, and the ability to operate at very high frequencies. Experimental results showed excellent agreement with theoretical predictions, confirming the effectiveness of the proposed design. This work illustrates the growing trend of integrating photonic and microwave technologies to overcome the limitations of traditional electronic filters and paves the way for the development of high-performance, tunable RF components. [28]

Wan et al. developed multi-band band-pass filters using square-ring resonator structures, aiming to address the increasing demand for multi-frequency operation in modern wireless communication systems. The proposed design utilizes the geometric properties of square-ring resonators to support multiple resonant modes within a single compact structure. By carefully adjusting the dimensions and coupling mechanisms of the resonator, the authors were able to achieve multiple passbands with high selectivity and low insertion loss. The study also explored the impact of various design parameters, such as coupling gap, resonator size, and substrate properties, on the filter performance. The results demonstrated that the proposed filter could operate efficiently across multiple frequency bands, making it suitable for applications such as multi-band wireless communication, satellite systems, and IoT devices. Additionally, the compact size and planar configuration allow easy integration into printed circuit boards, reducing overall system complexity. The authors also highlighted the importance of optimizing resonator geometry to achieve desired performance characteristics. This work provides valuable insights into the design of multi-band filters and demonstrates the effectiveness of square-ring resonators in achieving compact and high-performance filtering solutions. [29]

Ren et al. introduced a dual-band differential band-pass filter based on ring resonator structures, focusing on improving signal integrity and reducing noise in high-frequency communication systems. The differential configuration allows the filter to reject common-mode signals, which are a major source of interference in RF

systems. By utilizing ring resonators, the authors achieved dual-band operation with high selectivity and low insertion loss. The study included a detailed analysis of the design parameters, such as resonator dimensions, coupling coefficients, and substrate properties, to optimize filter performance. The authors demonstrated that the proposed filter exhibited excellent noise suppression and improved signal quality compared to conventional single-ended filters. Experimental validation confirmed the accuracy of the design and showed good agreement with simulation results. The compact size and high performance make the filter suitable for applications in modern wireless communication systems, including 5G and high-speed data transmission. This work highlights the importance of differential filter configurations in achieving high-performance RF systems and provides a strong foundation for future research in this area. [30]

Luo et al. proposed a dual-band band-pass filter using ring resonator structures, focusing on achieving efficient operation across multiple frequency bands while maintaining compact size and high performance. The design utilizes the inherent resonant properties of ring resonators to support two distinct frequency bands within a single structure. The authors demonstrated that by carefully controlling the coupling between resonators and adjusting geometrical parameters, it is possible to achieve independent tuning of the passbands. The proposed filter exhibited low insertion loss, high selectivity, and stable performance across both frequency bands. The study also analyzed the impact of fabrication tolerances on filter performance, highlighting the importance of precise design and manufacturing processes. The compact and efficient design makes it suitable for applications in modern wireless communication systems, where multi-band operation is essential. This work contributes to the development of advanced filter designs that can meet the increasing demands of multi-frequency communication systems. [31]

Kim and Chang developed an ultra-wideband (UWB) band-pass filter using ring resonator structures, targeting high-speed wireless communication systems that require large bandwidth and minimal signal distortion. The fundamental objective of their work was to overcome the limitations of conventional narrowband filters by achieving a significantly wider passband while maintaining good selectivity and low insertion loss. The proposed design utilizes the inherent resonance characteristics of

ring resonators, where multiple resonant modes are carefully controlled and combined to produce a continuous wideband response. The authors placed significant emphasis on optimizing the coupling mechanisms between the input/output feed lines and the resonator, as well as the internal coupling within the resonator structure, to achieve uniform transmission across the desired frequency range. Additionally, impedance matching techniques were applied to ensure minimal reflection loss throughout the band. The study also addressed the issue of spurious responses and harmonics, which are common challenges in UWB filter design, by carefully engineering the resonator geometry. Both simulation and experimental results demonstrated excellent agreement, confirming that the proposed filter achieved wide bandwidth, low insertion loss, and sharp roll-off characteristics. The compact planar structure further enhances its suitability for integration into modern RF and microwave circuits. This work is particularly important for applications such as ultra-wideband communication, radar systems, and high-speed data transmission, where efficient spectrum utilization and signal integrity are critical. Overall, the study highlights the effectiveness of ring resonator-based designs in achieving high-performance UWB filters and provides a strong foundation for further research in wideband filtering technologies. [32]

Patel et al. proposed a band-stop filter using ring resonator structures with the primary aim of effectively suppressing unwanted frequency bands that cause interference in communication systems. The design is based on the principle of resonance, where the ring resonator introduces a notch in the frequency response at specific frequencies corresponding to its resonant modes. The authors focused on achieving high selectivity and deep attenuation in the stopband region while maintaining minimal impact on the passband performance. A detailed parametric study was conducted to analyze the influence of key design parameters such as resonator dimensions, coupling gap, substrate characteristics, and feedline configuration on the overall filter performance. The results indicated that precise control of these parameters allows accurate tuning of the stopband frequency and bandwidth. The proposed filter demonstrated strong rejection characteristics, making it highly effective in eliminating interference from adjacent channels or unwanted signals in wireless communication systems. Additionally, the compact

planar design ensures ease of fabrication and compatibility with standard printed circuit board technologies. The authors also highlighted the potential applications of the filter in systems such as Wi-Fi, satellite communication, and radar, where interference suppression is crucial for maintaining signal quality. This work contributes significantly to the development of efficient and compact band-stop filters and demonstrates the practical applicability of ring resonator structures in modern RF systems. [33]

Çinar and Biçer developed a compact and efficient band-stop filter based on split-ring resonator (SRR) structures, focusing on achieving strong rejection of unwanted frequencies while maintaining simplicity and cost-effectiveness in design. The proposed filter utilizes the unique electromagnetic properties of split-ring resonators, which exhibit strong resonance behavior and can be engineered to produce sharp stopband characteristics. The authors designed the SRR structure to create a highly localized electromagnetic field, which enhances the interaction between the resonator and the incident signal, resulting in effective attenuation at the desired frequency. The study included a comprehensive analysis of design parameters such as resonator size, gap spacing, substrate material, and feedline configuration, demonstrating how these factors influence the stopband performance. The results showed that the filter achieved high selectivity, deep attenuation, and low insertion loss outside the stopband region. One of the key advantages of the proposed design is its compact size, which makes it suitable for integration into modern wireless devices where space is a critical constraint. Furthermore, the simple planar structure allows easy fabrication using conventional PCB techniques, reducing manufacturing complexity and cost. The authors also emphasized the potential applications of the filter in various RF systems, including mobile communication, radar, and electromagnetic interference suppression. This work highlights the effectiveness of split-ring resonator-based designs in developing compact and high-performance band-stop filters and contributes to the advancement of modern microwave filter technologies. [34]

CHAPTER 3

INTRODUCTION TO FILTERS

3.1 Introduction

In today's wireless communication era, filters are used efficient data transmission. Filters are widely used due to their compact size, low cost, and ease of fabrication. This work presents a U-shaped coupled with T-fed line band stop filter using the U-shaped Resonator technique to enhance bandwidth and radiation efficiency. A partial ground plane is incorporated to improve impedance matching and overall performance. The coupled U- shaped slots on the Substrate further optimizes radiation characteristics for advanced RF and S-band applications. Designed for the 4 GHz frequency range, the filter is implemented on an FR4 dielectric substrate for cost- effectiveness and ease of integration. Its improved radiation properties make it suitable for wireless and IoT applications. The optimized structure enhances performance in modern RF and cognitive radios. This research contributes to next-generation wireless communication with an efficient and compact filter design.

3.2 Characteristics of Filter

Filter possess several key characteristics that determine their performance in wireless communication systems. These parameters influence signal strength, directionality, efficiency, and overall communication reliability. Understanding these characteristics is essential for designing and selecting antennas for specific applications.

3.2.1 Radiation Pattern

- The directional distribution of radiated power from an antenna in space.
- Represented in 2D or 3D plots showing the intensity in different directions.

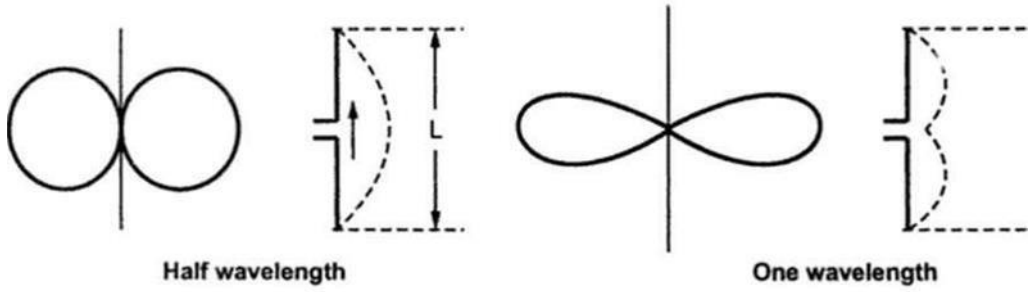


Figure 3.1: Radiation Pattern

3.2.2 Gain

- The measure of how effectively an antenna directs power in a specific direction compared to an isotropic radiator.

$$G = \eta D \qquad \text{eq. (1)}$$

- In decibels:

$$G(dB) = 10log10G \qquad \text{eq. (2)}$$

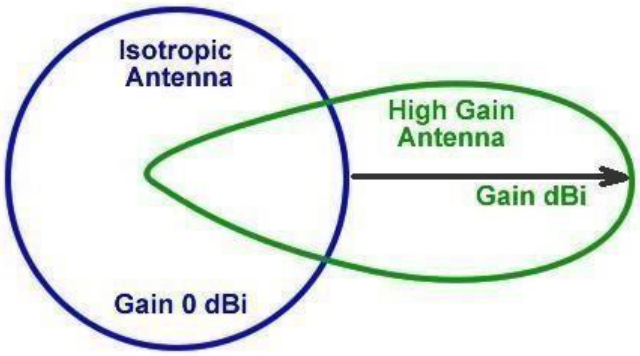


Figure 3.2: Gain

3.2.3 Directivity (D)

- The ability of an antenna to focus energy in a specific direction.

$$D = \frac{4\pi Umax}{Prad} \qquad \text{eq. (3)}$$

- In decibels:

$$D\ (dB) = 10\ log\ 10D \qquad \text{eq. (4)}$$

3.2.4 Bandwidth (BW)

- The frequency range over which the antenna maintains efficient performance.

$$BW = f_{high} - f_{low}$$

- Percentage bandwidth:

$$BW(\%) = \frac{f_{high} - f_{low}}{f_{center}} \times 100$$

3.2.5 Polarization

- The orientation of the electric field vector of the radiated wave.
- Can be linear, circular, or elliptical.

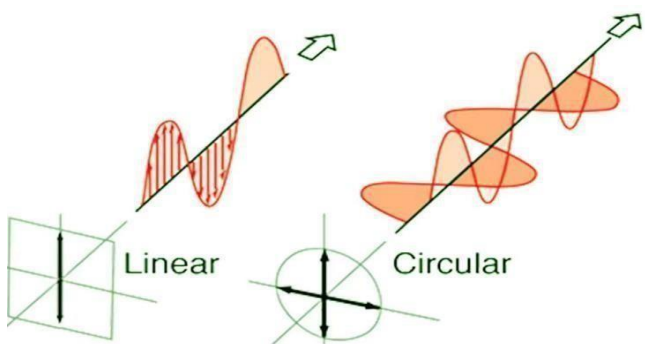


Figure 3.4: Polarization

3.2.6 Impedance Matching

- Ensuring the antenna's impedance (Z_A) matches the transmission line (Z_O) for minimal reflection.

$$\Gamma = \frac{Z_A - Z_O}{Z_A + Z_O} \qquad \text{eq. (5)}$$

- Ideal condition: $Z_A = Z_O$ for maximum power transfer.

3.2.7 VSWR (Voltage Standing Wave Ratio)

Indicates how well the filter is impedance-matched to the feedline.

$$VSWR = \frac{1+|\Gamma|}{1-|\Gamma|} \qquad \text{eq. (6)}$$

3.2.8 Return Loss (RL)

Measures power reflection due to impedance mismatch.

Higher return loss indicates better impedance matching.

$$RL(dB) = -20log10\,|\,\Gamma\,| \qquad \text{eq. (7)}$$

3.3 Types of filters

Filter play a crucial role in wireless communication systems by transmitting and receiving electromagnetic waves. Depending on their applications, structure ,and operating frequency, filter are classified into various types. In the context of biomedical and wearable applications, selecting the appropriate filter type is essential to ensure efficiency, reliability, and minimal interference. Below are some common types of filters:

Low-pass filter (LPF): It is designed to allow signals with frequencies lower than a specified cutoff frequency to pass through while attenuating higher frequencies. It is widely used in applications such as noise reduction, signal smoothing, and anti-aliasing in digital systems. The performance of an LPF is characterized by its cutoff frequency, roll-off rate, and passband ripple. In practical circuits, low-pass filters can be implemented using passive components like resistors, capacitors, and inductors, or active components such as operational amplifiers. In RF and microwave systems, LPFs are also used to suppress unwanted harmonics and ensure signal integrity.

High-pass filter (HPF): It performs the opposite function of a low-pass filter by allowing signals above a certain cutoff frequency to pass while attenuating lower frequencies. High-pass filters are commonly used to remove low-frequency noise, DC components, and baseline drift in signals. They are essential in audio processing systems to eliminate rumble and improve clarity, as well as in communication systems where only high-frequency components carry useful information. Like LPFs, HPFs can be realized using both passive and active circuit elements, and their design focuses on achieving sharp cutoff characteristics and minimal signal distortion.

Band-pass filter (BPF): It allows a specific range of frequencies to pass while attenuating frequencies outside this range. It is characterized by two cutoff frequencies: a lower cutoff frequency and an upper cutoff frequency, which define the bandwidth of the filter. Band-pass filters are widely used in wireless communication systems, radio receivers, and signal processing applications where it is necessary to isolate a particular frequency band. The quality factor (Q-factor) of a band-pass filter determines its selectivity, with higher Q indicating a narrower and more selective passband.

In microwave engineering, band-pass filters are often implemented using resonators, transmission lines, or cavity structures.

Wideband band-pass filter: It is designed to allow a broad range of frequencies between the lower and upper cutoff limits. It has a relatively large bandwidth and is typically used in applications where a wide spectrum of signals needs to be transmitted or received, such as broadband communication systems and RF front-end circuits. These filters are generally simpler in design and less sensitive to component variations, but they offer lower selectivity compared to narrowband filters.

Narrowband band-pass filter: It permits only a very small range of frequencies around a center frequency. It has a high quality factor (Q), indicating high selectivity and sharp response characteristics. Narrowband filters are widely used in applications such as radio receivers, wireless communication systems, and signal processing where precise frequency selection is required. However, they require careful design and tuning, as even small variations in components can significantly affect performance

Band-stop filter (BSF): It is also known as a notch filter, is designed to reject a specific range of frequencies while allowing all other frequencies to pass. This type of filter is particularly useful for eliminating unwanted interference or noise at certain frequencies, such as power line interference in electronic systems. The rejected frequency range is called the stopband, and the effectiveness of the filter is measured by its attenuation level and bandwidth. Band-stop filters are widely used in RF and microwave applications, including radar, satellite communication, and cognitive radio systems. Advanced designs, such as tunable band-stop filters, enable dynamic adjustment of the stopband frequency and bandwidth, making them highly suitable for modern adaptive communication system

Wideband band-stop filter: It is designed to suppress a broad range of frequencies around the center frequency. It has a relatively large bandwidth, meaning the difference between the upper and lower cutoff frequencies is high. These filters are typically used in applications where a wide range of unwanted signals or noise needs to be eliminated, such as in electromagnetic interference (EMI) suppression and broadband communication systems. Wideband band-stop filters are generally easier to design but may exhibit lower selectivity compared to narrowband filters.

Narrowband band-stop filter, also known as a notch filter, is designed to reject a very small and specific frequency range. It has a high quality factor (Q) indicating sharp selectivity and precise frequency rejection. Narrowband filters are commonly used to eliminate single-frequency interference, such as power line noise (50/60 Hz) in electronic circuits, audio systems, and medical instruments. These filters require precise design and tuning to achieve high attenuation at the desired frequency without affecting adjacent frequencies.

All-pass filter (APF) is a unique type of filter that allows all frequency components of a signal to pass through with equal amplitude while altering only the phase of the signal as a function of frequency. Unlike other filters such as low-pass, high-pass, band-pass, and band-stop filters, which affect the magnitude response, an all-pass filter maintains a constant gain across all frequencies and modifies only the phase response. This characteristic makes it especially useful in applications where phase alignment, delay control, or signal timing adjustment is required without affecting signal strength.

All-pass filters are commonly used in communication systems, audio signal processing, and control systems to correct phase distortion introduced by other components. For example, in multi-stage filtering systems, different filters may introduce varying phase shifts, leading to signal distortion. An all-pass filter can be used to compensate for this phase variation and restore signal integrity. Additionally, all-pass filters are used to implement phase shifters, group delay equalizers, and in the design of oscillators and feedback systems.

3.4 Band pass fliter

3.4.1 Introduction

A bandpass filter (BPF) is a crucial frequency-selective component used in RF and microwave communication systems to allow a specific range of frequencies to pass while attenuating frequencies outside this range. The frequency range that is allowed to pass is known as the passband, which is defined by the lower cutoff frequency (fL) and upper cutoff frequency (fH). The midpoint of this range is called the center frequency (fc), where the filter exhibits maximum transmission efficiency.

In modern wireless communication systems, such as ISM band applications and 5G technologies, bandpass filters play a vital role in ensuring efficient signal transmission and reception. They help eliminate unwanted signals, reduce noise, and prevent interference between adjacent channels. In this project, the designed bandpass filter operates around 4 GHz center frequency with a passband range of 3.5 GHz to 4.5 GHz, making it suitable for wireless communication applications.

The proposed design uses a compact microstrip resonator-based structure, which enhances filter performance while maintaining a small physical size. Unlike conventional filters fabricated on FR-4 substrates, which suffer from higher dielectric losses and poor selectivity, this design utilizes Rogers RT/Duroid 5880 substrate, which provides low dielectric loss and improved high-frequency performance. Additionally, simulation using ANSYS HFSS ensures accurate modeling of electromagnetic behavior, enabling optimization of return loss, insertion loss, and bandwidth.

Thus, the bandpass filter designed in this work provides improved efficiency, better impedance matching, and enhanced signal selectivity, making it suitable for RF front-end systems, IoT devices, and modern wireless communication applications.

3.4.2 Structure of Bandpass filter

The structure of the proposed bandpass filter is based on a compact microstrip resonator configuration, where resonating elements are designed to achieve the desired frequency response. In microstrip technology, the filter is implemented on a dielectric substrate with a conducting patch on one side and a ground plane on the other side. The electromagnetic waves propagate through the microstrip line, and the resonator elements create selective frequency transmission.

In this design, the resonator structure is carefully modified to enhance coupling between elements and achieve a compact layout. The coupling between resonators determines the bandwidth of the filter, while the physical dimensions of the resonator control the center frequency. By adjusting parameters such as resonator length, width, and spacing, the desired passband characteristics are obtained.

The use of Rogers RT/Duroid 5880 substrate significantly improves performance due to its low dielectric constant and minimal loss tangent, which ensures reduced signal attenuation and improved transmission efficiency. The filter is excited using a microstrip feedline, which provides uniform signal distribution and ensures compatibility with planar circuits.

Furthermore, the structure is optimized to achieve low return loss (S11 < -10 dB) and high transmission (S21) within the passband, indicating efficient signal propagation. The compact nature of the design makes it suitable for integration into modern wireless devices where space and performance are critical factors.

3.4.3 Advantages of Bandpass filter

• Allows only the desired frequency band to pass while rejecting unwanted signals.
• Improves signal quality by reducing noise and interference in communication systems.
• Provides high selectivity and efficient frequency discrimination.
• Compact microstrip design enables easy integration into RF circuits.
• Low dielectric loss using Rogers substrate improves transmission efficiency.
• Suitable for ISM band, 5G, and IoT applications.
• Enhances overall system performance in wireless communication systems.

3.4.4 Disadvantages of Bandpass filter

• Design complexity increases when high selectivity and narrow bandwidth are required.
• Requires precise dimension control for accurate frequency response.
• Fabrication errors may affect performance at high frequencies.
• Higher cost due to the use of advanced substrates like Rogers materials.
• Performance is sensitive to variations in substrate properties and environmental conditions.
Limited tuning range in conventional designs without using additional components.

3.4.5 Feeding Techniques

Microstrip Line Feeding:
This is the primary feeding technique used in the proposed design. It involves a conducting strip printed on the dielectric substrate, which excites the resonator structure. It is simple, cost-effective, and widely used in planar RF circuits.

Coplanar Waveguide (CPW) Feeding:
In CPW feeding, both the signal line and ground plane are placed on the same layer, providing better integration with active components and reduced radiation loss.

Coaxial Feeding:
This method uses a coaxial connector to excite the filter, providing good shielding and minimal interference, mainly used in measurement setups.

Waveguide Feeding:

Used in high-frequency applications, offering low loss and high power handling capability, but it is bulky and not suitable for compact designs.

3.4.6 Applications of Bandpass filter

- Used in wireless communication systems for selecting desired frequency bands.
- Applied in RF front-end modules and transceivers.
- Essential in ISM band applications such as Wi-Fi and Bluetooth.
- Used in IoT devices for efficient signal processing.
- Applied in radar and satellite communication systems.
- Used in signal processing to eliminate unwanted frequency components.
- Important in 5G communication systems for spectrum management.

3.4.7 Design Considerations for Bandpass fliter

Center Frequency Selection: The desired center frequency must be accurately defined, as it determines the midpoint of the stopband where maximum attenuation occurs.

Bandwidth:

The width of the stopband (difference between upper and lower cutoff frequencies) should be selected based on application needs.

Quality Factor(Q):

A higher Q-factor provides sharper selectivity and narrower bandwidth, which is important for precise frequency rejection.

Insertion Loss: The filter should have minimal insertion loss in the passband to ensure efficient signal transmission outside the stopband.

Return Loss & Impedance Matching:

Proper impedance matching is required to reduce reflections and improve overall performance.

3.5 Conclusion

In this chapter, a detailed study of the bandpass filter has been presented with specific emphasis on the proposed compact microstrip resonator-based design. The fundamental concepts of bandpass filters, including their structure, working principle, advantages, disadvantages, feeding techniques, applications, and design considerations, have been thoroughly discussed.

The proposed bandpass filter is designed to operate around a center frequency of approximately 4 GHz with a passband ranging from 3.5 GHz to 4.5 GHz, making it suitable for modern wireless communication applications. The use of a microstrip resonator structure enables compact size and ease of fabrication, while the selection of Rogers RT/Duroid 5880 substrate significantly improves performance by reducing dielectric losses and enhancing signal transmission efficiency.

Key design parameters such as bandwidth, return loss, insertion loss, and impedance matching have been carefully considered to achieve optimal filter performance. The microstrip feeding technique ensures effective signal excitation and compatibility with planar RF circuits. Additionally, the design provides good selectivity and minimal signal distortion within the passband, which is essential for high-frequency applications.

Overall, the proposed bandpass filter demonstrates improved performance characteristics compared to conventional designs, making it suitable for applications in wireless communication systems, RF front-end modules, and IoT devices. The detailed understanding gained from this chapter forms a strong foundation for the subsequent simulation and implementation of the filter using ANSYS HFSS, which will be discussed in the following chapters.

CHAPTER 4

ANSYS HFSS SOFTWARE

4.1 Introduction

This chapter focuses on the design, simulation, and analysis of a bandpass filter using ANSYS HFSS (High-Frequency Structure Simulator). The HFSS software is a powerful tool for solving full-wave electromagnetic problems, ensuring accurate antenna performance analysis.The complete workflow involves:

- Setting up the HFSS workspace.
- Defining material properties.
- Creating the antenna structure.
- Configuring boundary conditions and excitation.
- Running the simulation.
- Extracting key performance parameters.

4.2 HFSS Software Overview

Ansys HFSS is a leading electromagnetic simulation tool for designing high-frequency components like antennas, waveguides, and RF devices. Using the Finite Element Method (FEM), it provides accurate full-wave solutions, making it essential for antenna engineers and researchers.

Figure 4.1: Ansys HFSS

4.2.1 Key Capabilities of HFSS in Electromagnetic Analysis

Full-Wave Electromagnetic Simulation HFSS performs three-dimensional full-wave analysis, solving Maxwell's equations to predict electromagnetic field behavior with high precision. This ensures accurate evaluation of antenna and filter parameters such as radiation efficiency, impedance matching, and polarization characteristics.

- **Full-Wave Electromagnetic Simulation**

  HFSS performs three-dimensional full-wave analysis, solving Maxwell's equations to predict electromagnetic field behavior with high precision. This ensures accurate evaluation of filter parameters such as radiation efficiency, impedance matching, and polarization characteristics.

- **S-Parameter Analysis (Scattering Parameters)**

  The software enables the extraction of S-parameters (S11, S12, etc.), which are crucial for understanding an antenna's impedance matching and return loss performance. The S11 parameter, in particular, determines the reflection coefficient, helping engineers optimize antenna design for minimal power loss.

- **Radiation Pattern Computation**

  HFSS computes both 2D and 3D radiation patterns, allowing visualization of an antenna's directivity and beamforming characteristics. This is essential in evaluating the performance of microstrip patch antennas and filters in different orientations and environments.

- **Field Distribution Analysis**

  The software provides detailed insights into electric and magnetic field distributions, enabling designers to assess current flow on the patch, fringing fields, and radiation efficiency. This helps in fine-tuning antenna geometry for enhanced performance.

- **Automatic Adaptive Meshing**

  HFSS employs adaptive meshing techniques to optimize simulation accuracy while reducing computational overhead. The meshing process refines critical regions of the design, ensuring precise results for return loss, gain, and impedance bandwidth.

- **Integration of Material and Boundary Conditions**

  The software allows the inclusion of various dielectric materials, boundary conditions, and lumped elements to simulate real-world antenna scenarios. This ensures the accurate modelling of microstrip antennas with Felt substrates, ground plane modifications, and feeding techniques.

- **Optimization and Parametric Analysis**

  HFSS provides design optimization tools that help adjust antenna and filter parameters, such as patch dimensions, feedline width, slot size, and ground plane configurations, to achieve the best possible performance in terms of gain, bandwidth, and return loss.

- **Antenna Array and Multi-Port Simulation**

  The tool supports multi-element antenna array simulation, enabling engineers to analyze MIMO (Multiple-Input Multiple Output) and phased array systems for advanced wireless communication applications.

4.3 HFSS Software Setup:

ANSYS HFSS (High-Frequency Structure Simulator) is a powerful tool for designing and analyzing high-frequency electromagnetic structures. This section covers the complete setup process from launching HFSS to defining simulation parameters. It includes creating the antenna model, assigning boundary conditions, setting up excitation, and configuring the solution setup. Proper meshing and frequency setup ensure accurate simulation results. Once configured, the simulation is run to analyze various antenna performance metrics.

4.3.2 Launching HFSS and Creating a New Project

- **Open ANSYS HFSS:**

  Click on HFSS 3D Layout Design from the ANSYS Electronics Desktop.

  Select File → New Project to start a new simulation.

- **Define the Solution Type:**

  Go to HFSS → Solution Type and choose Driven Modal for wave-based Analysis.

  Click OK to confirm.

4.3.3 Creating the Antenna Model

- **Defining the Substrate**

  - **Insert a New Box:**

    Click Draw → Box , and place it on the X-Y plane.

    Right-click and select Properties.

    Set Length, Width, and Height based on the substrate dimensions.

- **Assign Material to the Substrate:**

  - Open the Materials library.

- Select Felt ($\varepsilon_r = 1.3$, $\tan \delta = 0$) or define a custom dielectric.
- Assign the material to the substrate geometry.

- **Designing the Patch**
  - **Draw a Cylindrical Patch:**

    Click Draw → Cylinder and select the X-Y plane.

    Define the radius based on the resonant frequency calculation.

  - **Create Slot on the Patch:**

    Click Draw → Cylinder/Circle and place the slot on the patch.

    Use Boolean → Subtract to remove the slot area.

4.3.4 Defining the Ground Plane

- **Draw the Ground Plane:**

  Create a Box beneath the substrate (same dimensions as the substrate).

  Assign PEC (Perfect Electric Conductor) as the material.

4.3.5 Configuring Boundary Conditions and Excitation

- **Applying Radiation Boundary**
  - **Create an Air Box:**

    Draw a large box surrounding the antenna ($\lambda/4$ spacing from all sides).

    Assign vacuum as the material.

- **Set Radiation Boundary:**

  Right-click the air box → Assign Boundary → Radiation.

- **Defining Wave Port Excitation**

  - **Create a Feed Line:**

    Draw a thin microstrip line connecting the patch.
    Assign PEC material.

- **Define the Wave Port:**

  Select the microstrip feed end → Assign Excitation → Wave Port.
  Set the port size as width, height where h is the substrate thickness.

4.3.6 Setting Up Simulation Parameters

- **Mesh Generation**

  Go to HFSS → Mesh → Assign Mesh.
  Select Fine Mesh for accurate results.

- **Frequency Setup**

  Go to HFSS → Analysis Setup → Add Solution Setup.
  Set Start Frequency and End Frequency (based on the operating range).
  Click Adaptive Meshing to refine the accuracy.

4.4 Running the Simulation

Once the HFSS model setup is complete, the simulation process begins. The designed antenna structure, boundary conditions, and excitation settings are analyzed for accuracy. The simulation is executed by selecting HFSS → Analyze All, which computes the electromagnetic response of the antenna. The solver iterates through adaptive meshing to refine results and ensure convergence. If errors occur, adjustments to mesh settings, boundary conditions, or feed placement may be required to optimize performance.

- Click HFSS → Analyze All to start the simulation.
- Wait for the results to converge.

4.5 Extracting A Performance Parameters

After running the simulation, key antenna parameters are analyzed to evaluate its performance. The results provide insights into return loss, impedance matching, radiation characteristics, and efficiency. Various plots and reports are generated in HFSS to study these parameters. The following sections detail the extraction of essential performance metrics, including Return Loss (S11), VSWR, Radiation Pattern, Gain, Directivity, Isolation, ECC, CCL, and DG.

4.5.1 Return Loss (S11) and Bandwidth

Return loss (S11) is a critical parameter used to evaluate how much power is reflected back from the antenna due to impedance mismatch. Ideally, an antenna should have minimal reflections, ensuring maximum power transfer. The return loss is measured in decibels (dB) and is considered optimal when it is below -10 dB, as this signifies that at least 90% of the transmitted power is accepted by the antenna.

To analyze return loss in HFSS, follow these steps:

- Navigate to Results → Create Report →Rectangular Plot.
- Select S11 Parameter vs. Frequency to generate the return loss curve.
- Identify the resonant frequencies where S11 < -10 dB, indicating strong impedance matching.

The bandwidth of an antenna is also determined using the return loss curve, where the frequency range over which S11 remains below -10 dB defines the operational bandwidth. A wider bandwidth ensures that the antenna can operate efficiently over multiple frequency ranges, making it suitable for broadband communication applications.

4.5.2 VSWR (Voltage Standing Wave Ratio) Analysis

Voltage Standing Wave Ratio (VSWR) is another essential parameter for evaluating the impedance matching between the antenna and the feedline. A lower VSWR value indicates better power transmission with minimal reflections, leading to higher efficiency.

For an efficient antenna, VSWR should ideally be less than 2, meaning that the reflected power is negligible. A VSWR close to 1 signifies near-perfect matching, where almost all the power is radiated. The VSWR is calculated using the equation:

To plot VSWR in HFSS, follow these steps:

- Navigate to Results → Create Report → VSWR.
- Generate the VSWR plot to examine the matching characteristics at the resonant frequencies.
- Ensure that VSWR < 2 at the primary resonant frequencies, indicating optimal performance.

If the VSWR exceeds 2, impedance tuning techniques such as adjusting the feed location, adding matching networks, or modifying antenna geometry can be applied to improve the antenna's efficiency.

4.5.3 Radiation Pattern Analysis

The radiation pattern of an antenna describes the distribution of radiated energy in space. It helps determine the antenna's directionality, beamwidth, and efficiency in transmitting signals. The radiation pattern is usually analyzed in both 2D (E-plane and H-plane) and 3D views to understand its spatial emission characteristics.

To visualize the radiation pattern in HFSS, follow these steps:

- Navigate to Results → Create Report → Far-Field Radiation.
- Choose either 2D plots (E-plane, H-plane) or 3D plots for a complete visualization.
- Observe the main lobes, side lobes, and back lobes to assess directivity and beam shaping.
- The E-plane radiation pattern represents the electric field distribution, showing how directional the antenna is.
- The H-plane radiation pattern depicts the magnetic field distribution, often indicating omnidirectional behavior.

4.5.4 Gain and Directivity Evaluation

Gain and directivity are crucial performance parameters that define the effectiveness of an antenna in radiating power in a specific direction.

- Directivity refers to how concentrated the radiated power is in a particular direction compared to an isotropic radiator.
- Gain represents the efficiency of power transmission in a given direction, considering both directivity and radiation efficiency.

To compute Gain and Directivity in HFSS, follow these steps:

- Navigate to Results → Compute Gain & Directivity.
- Observe the gain values at resonant frequencies.
- If the gain is greater than 5 dB, it indicates efficient radiation with minimal losses.

CHAPTER 5

DESIGN METHODOLOGY

5.1 Introduction

The design and development of a bandpass filter require careful consideration of several parameters, including substrate material, resonator geometry, feed mechanism, and coupling structure. In this chapter, the step-by-step implementation of the compact microstrip resonator-based bandpass filter using ANSYS HFSS is presented. HFSS is a widely used electromagnetic simulation tool that employs the Finite Element Method (FEM) to accurately analyze high-frequency RF and microwave components such as filters and antennas.

The proposed bandpass filter is designed to operate around 4 GHz, with a passband ranging approximately from 3.5 GHz to 4.5 GHz. The design focuses on achieving improved bandwidth, low insertion loss, and enhanced impedance matching. Unlike antenna-based structures, the filter relies on resonator coupling and transmission line behavior to achieve frequency selectivity.

The design follows an iterative approach, where the resonator structure, coupling gaps, and feedline configuration are progressively optimized to achieve the desired S-parameter response (S11 and S21). This systematic approach ensures that the final design meets the requirements of modern wireless communication systems.

5.2 Design Procedure

This section presents the step-by-step methodology for designing the microstrip bandpass filter in HFSS. The process includes substrate selection, resonator design, feedline implementation, coupling optimization, and simulation-based refinement.

5.2.1 Substrate Selection

The substrate plays a critical role in determining the performance of the bandpass filter, affecting parameters such as insertion loss, bandwidth, and resonant frequency

Parameter	Symbol	Value
Material	-	Rogers RT/Duroid 5880
Dielectric Constant	ε_r	2.2
Loss Tangent	tan δ	0.0009
Thickness	h	1 mm

Table I: Key Parameters of the Felt Substrate

The low dielectric constant and low loss tangent of the substrate help in achieving high efficiency and reduced signal loss, making it suitable for high-frequency applications such as RF filters.

5.2.2 Resonator Design

Unlike antennas, the bandpass filter uses resonators instead of radiating patches.

• **Define Resonator Geometry:**

The resonator dimensions are calculated based on the required center frequency (~4 GHz). The length of the resonator determines the resonant frequency.

• **Coupling Mechanism:**

The spacing between resonators controls the bandwidth of the filter. Stronger coupling results in wider bandwidth.

• **Parametric Optimization:**

HFSS parametric analysis is used to fine-tune resonator length, width, and spacing to achieve the desired S21 response.

- **Optimize Resonator Geometry:**

  A parametric analysis is conducted in HFSS to refine the slot dimensions.

  The slot radius is adjusted iteratively to achieve the best possible performance.

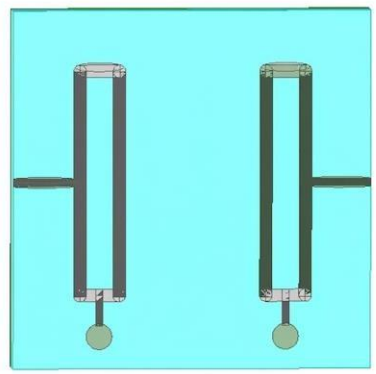


Figure 5.1: Initial resonator structure

5.2.3 Ground Plane Design

A full ground plane is implemented to provide stable impedance characteristics and consistent radiation performance for the antenna.

- **Define Ground Plane Dimensions:**

  The ground plane length (L_s) is maintained equal to the Resonator length to ensure structural symmetry and stable radiation characteristics.

  The ground plane width (W_s) is kept equal to the substrate width to ensure proper return loss characteristics.

- **Simulation-Based Optimization:**

  Multiple ground plane lengths are tested using HFSS simulations.

  The configuration that provides the best return loss (S11 < -10 dB) and VSWR < 2 is chosen.

- **Enhance Bandwidth:**

  The full ground plane helps provide a stable reference for the antenna and supports proper current flow.

  The ground plane improves current distribution, resulting in stable radiation characteristics and good antenna performance.

  By maintaining a full ground plane, the antenna achieves stable impedance behavior and reliable radiation performance, which is essential for consistent operation.

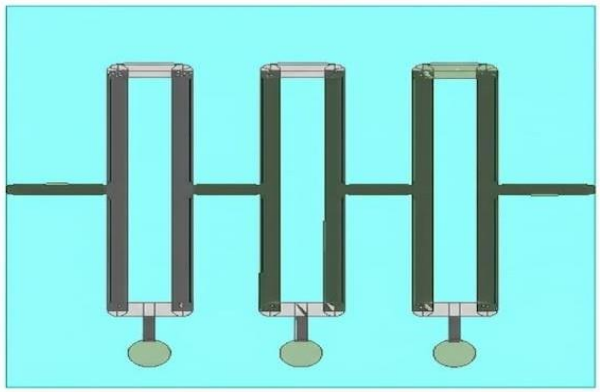


Figure 5.2: Extended resonator structure with multiple units

5.2.4 Feed Mechanism

The feed mechanism determines how efficiently power is transferred to the antenna. In this design, a microstrip line feed is used due to its low complexity, ease of integration, and efficient power transfer.

- **Design the Feed Line:**

  The width of the feed line is calculated using the 50Ω impedance matching formula.

  A well-matched feed line ensures that maximum power is transferred from the source to the antenna with minimal reflections

- **Optimize Feed Placement:**

  The feed point is adjusted iteratively to ensure low reflection loss and good return loss characteristics.

  The feed line is placed at an optimal distance to achieve resonant frequency matching.

- **Ensure Proper Power Transmission:**

  The microstrip feed minimizes power loss and ensures the patch is excited efficiently for optimal radiation performance.

  By optimizing the feed mechanism, the antenna maintains low return loss, stable impedance matching, and efficient excitation.

5.2.5 Slot Implementation

One of the most crucial aspects of this design is the implementation of slot, which play a key role in enhancing bandwidth and radiation efficiency.

- **Design Steps**

Determine Slot Location:

The rectangular slot is positioned symmetrically at the center of the patch.

The placement of slot alters the surface current distribution, leading to improved impedance matching.

Optimize Slot Size:

The radius (R) of the cylindrical slot is fine-tuned through parametric analysis in HFSS.

The slot dimensions are adjusted iteratively to achieve optimal performance.

Validate via Simulation:

HFSS simulations confirm that return loss (S11) is significantly improved, and VSWR remains below 2.

The presence of cylindrical slot enhances the dual band operation of the filter. By integrating cylindrical slot, the filter achieves better bandwidth performance, lower return loss, and higher efficiency.

5.3 Evolution of Filter Design in Three Stages

The proposed filter design underwent iterative modifications in three stages, refining its geometry, slot and ground plane structure to achieve optimal performance.

5.3.1 Basic Patch Filter Design

A cylindrical microstrip patch with no slots was initially designed.
The performance was evaluated, and the impedance bandwidth was found to be limited.

5.3.2 Introduction of Cylindrical Slot

A cylindrical slot is added to the center of the patch.
The slots enhanced impedance bandwidth and improved return loss.
The resonant frequency was slightly shifted, requiring finetuning.

5.3.3 Optimization of Ground Plane and Feed Mechanism

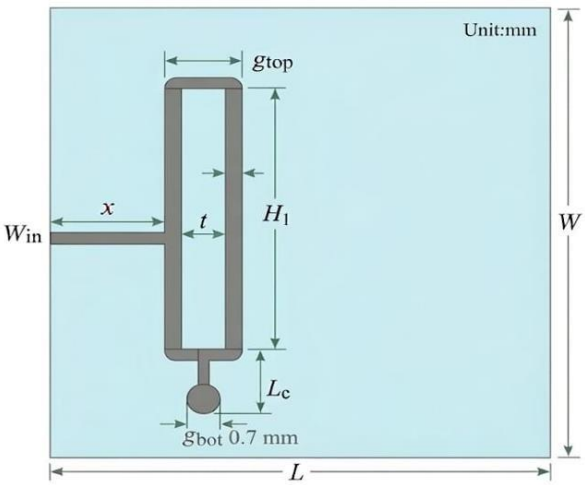
A full ground plane was implemented to further enhance bandwidth.
The feed position was optimized to achieve better impedance matching.
HFSS simulations confirmed improved return loss (S11 < -58dB) and VSWR < 2.

5.3.4 Final Design Outcome:

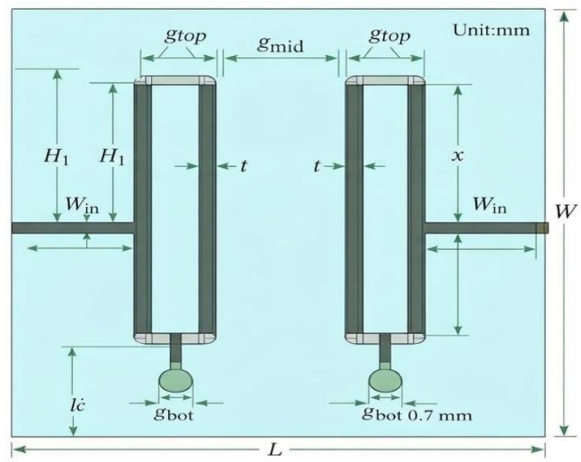
Dual-band operation achieved.

Enhanced impedance bandwidth.

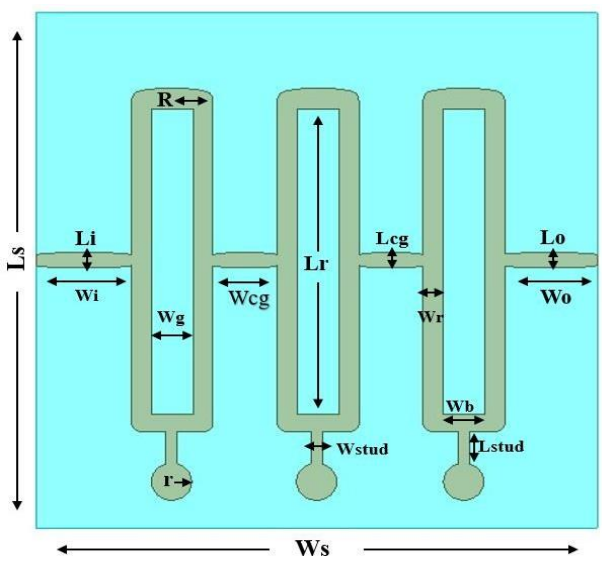
Lower return loss and improved radiation efficiency.



Stage 1



Stage 2



Stage 3

Figure 5.3: Various stages of the band pass filter

5.4 Finalized Filter Dimensions

The finalized filter dimensions are determined through extensive design calculations, simulations, and optimization using HFSS. Each dimension is carefully chosen to ensure optimal impedance matching, bandwidth enhancement, and radiation efficiency.

PARAMETER	SYMBOL	DIMENSIONS(mm)
Substrate Length	Ls	10
Substrate Width	Ws	10
Coupling Length	Lcg	0.7
Coupling Width	Wcg	1.15
Resonator Length	Lr	6.3
Resonator Width	Wr	0.35
Input Length	Li	1.05
Input Width	Wi	1.7
Output Length	Lo	1.05
Output Width	Wo	1.7
Thickness	T	1
Stud Length	Lstud	0.7
Stud Width	Wstud	0.2
Gap Width	Wg	1.11
Raidus(E)	R	0.68
Radius(C)	r	0.36

Table II: Proposed Filter Dimension

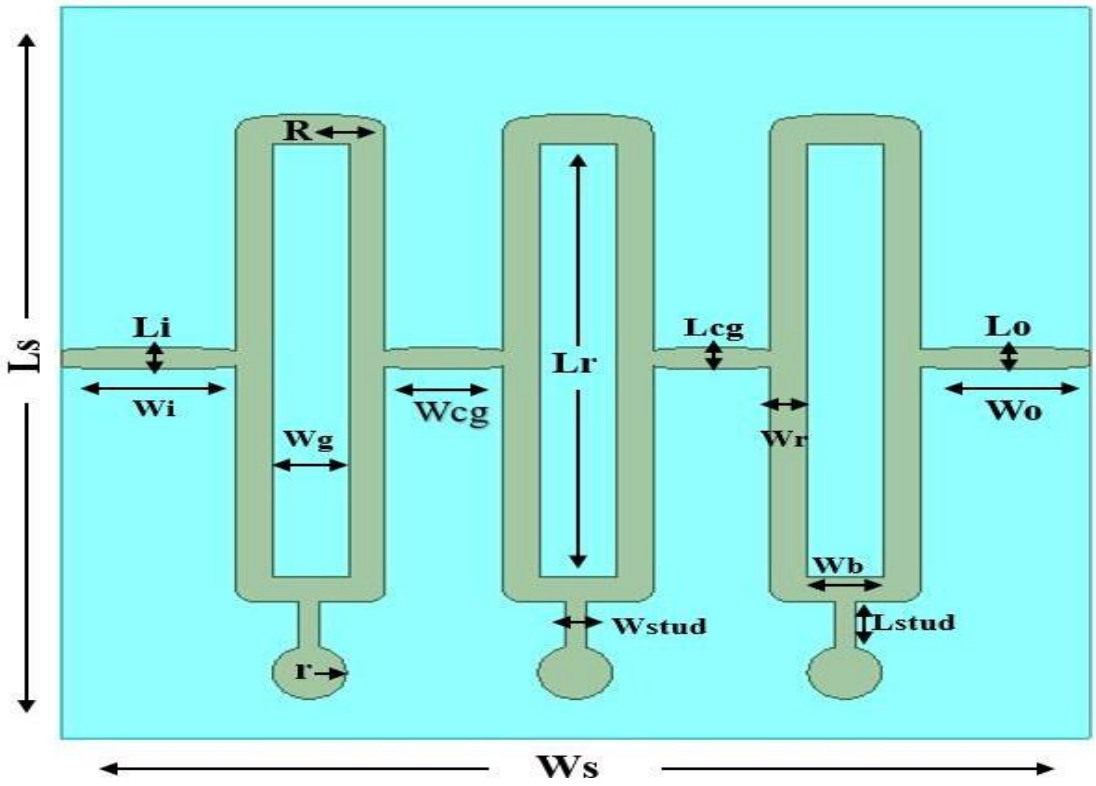


Figure 5.4: Proposed Filter Design

5.5 Rectangular Slot Implementation

The primary modification in the proposed filter is the introduction of cylindrical slot at the external edges of the patch. These slots play a crucial role in enhancing radiation characteristics, improving bandwidth, and achieving optimal impedance matching.

Key aspects of Cylindrical Slot Implementation:

- **Slot Placement:**

  A cylindrical slot is symmetrically introduced at the center of the circular patch.

- **Impact on Current Distribution:**

  The slot introduces additional current paths, affecting the resonant frequency and impedance matching, which helps in achieving an extended bandwidth.

- **Enhancement of Radiation Efficiency:**

  The slot alters the surface current distribution, leading to better radiation efficiency and improved gain.

  They enable a dual-band response, making the filter suitable for multiple frequency operations.

- **Reduction in Filter Size:**

  The introduction of cylindrical slot helps in reducing the overall filter size while maintaining high-performance characteristics.

  This is achieved without compromising parameters like gain, return loss, and radiation efficiency.

- **Optimization through HFSS Simulations:**

  The radius of the cylindrical slot was iteratively finetuned in HFSS to achieve the best frequency response.

  Parametric analysis was conducted to find the optimal slot dimensions that minimize reflection loss and maximize bandwidth.

5.6 Simulation Setup and Boundary Conditions

To ensure accurate and reliable electromagnetic simulations of the proposed filter, a well-structured simulation environment was implemented using Ansys HFSS. The simulation setup consists of excitation methods boundary conditions, and cylindrical slot implementation, each playing a critical role in achieving precise and realistic results.

5.6.1 Excitation Method – Wave Port Excitation

The Wave Port excitation method was employed to excite the microstrip patch antenna and filters effectively. This approach provides a highly accurate representation of the real- world feeding mechanism, ensuring proper impedance matching between the microstrip feedline and the filter structure.

Key aspects of the Wave Port Excitation setup:

Wave Port Assignment: The Wave Port is assigned at the feed line input, ensuring efficient signal propagation with minimal reflection and loss.

Impedance Matching: The port is calibrated to a characteristic impedance of 50 Ω, which is the standard impedance for RF and microwave applications.

S-Parameter Extraction: The Wave Port excitation facilitates the extraction of S-parameters such as S11 and S21, which are critical for evaluating

Return Loss (S11) – to assess how much power is reflected back.

Reflection Coefficient – to determine impedance matching.

Impedance Bandwidth – to verify the frequency range over which the filter operates effectively.

Minimization of Losses: This excitation method ensures efficient energy transfer from the feedline to the patch, minimizing unwanted losses

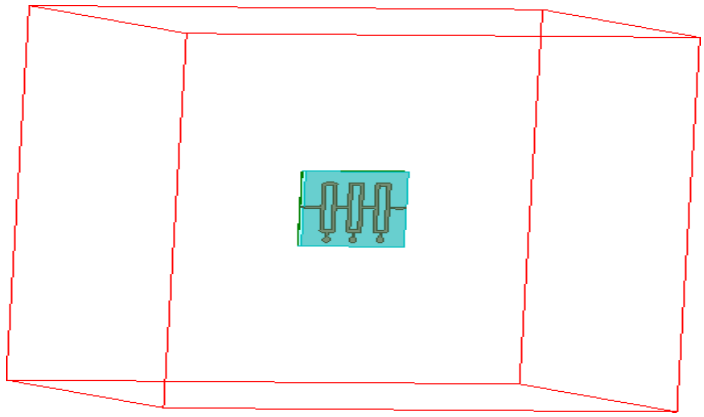


Figure 5.5: Excitation Port

5.6.2 Boundary Conditions – Radiation Boundaries

To simulate the filter's free-space behavior and eliminate unwanted reflections from the simulation domain, radiation boundaries were applied. These boundaries play a crucial role in achieving realistic electromagnetic wave propagation without distortion.

Key considerations for Radiation Boundary Setup:

- **Application of Radiation Boundary:** A radiation boundary was assigned to all outer regions surrounding the filter structure.
- **Reflection Minimization:** The boundary was positioned at a distance of at least λ/4 (quarter-wavelength) from the filter to ensure that reflections from simulation domain edges do not interfere with the results.
- **Far-Field Parameter Optimization:** Byapplying radiation boundaries, HFSS can accurately predict key far-field parameters, such as:
- **Radiation Patterns** – to analyze how the filter radiates in different directions.
- **Gain and Directivity** – to evaluate signal strength and focusing ability.

- **Efficiency** – to determine the proportion of input power successfully converted into radiated power.
- **Ensuring Accuracy in Open-Space Conditions:** The proper placement of radiation boundaries mimics the real-world free space operation of the filter, preventing artificial reflections that could distort simulation results

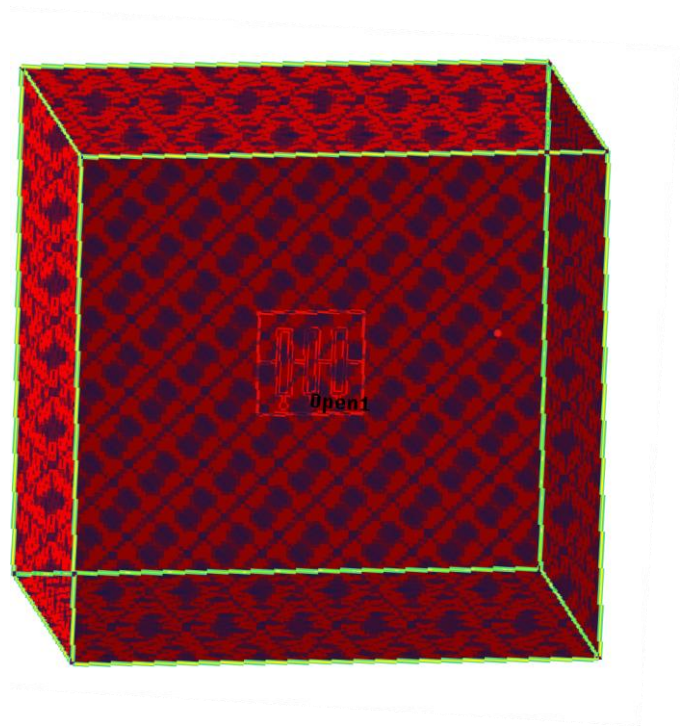


Figure 5.6: Boundary Condition

5.6.3 Performance Evaluation

The performance of the filter is assessed using Ansys HFSS simulations, focusing on key parameters such a return loss(S11), VSWR, bandwidth, gain, radiation efficiency, and far- field radiation patterns to ensure optimal operation within the desired frequency bands.

5.6.4 Return Loss (S11) Analysis

Return loss (S11) measures power reflection due to impedance mismatching. A value below -58 dB ensures efficient power transfer. The S11 vs. frequency plot confirms single- band operation with improved impedance matching.

5.6.5 Voltage Standing Wave Ratio (VSWR) Evaluation

VSWR indicates transmission efficiency, with values below 2 ensuring minimal reflection. HFSS simulations validate that the filter meets this criterion across both bands.

5.6.6 Bandwidth Enhancement

The cylindrical slot and partial ground plane enhance bandwidth. The S11 plot determines the frequency range where S11 remains below -58 dB, showing significant improvement over the initial design.

5.6.7 Gain and Radiation Efficiency

Gain (measured in dB) determines radiated energy direction, while radiation efficiency evaluates power conversion effectiveness. 3D radiation patterns in HFSS confirm improved gain and efficiency.

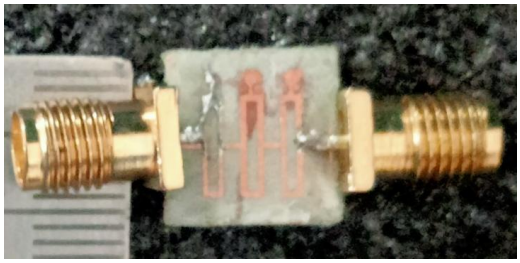
5.6.8 Far-Field Radiation Patterns

The far-field pattern assesses main lobe direction, half-power beamwidth (HPBW), and side lobe levels. The optimized design provides better directivity with an omni-directional pattern.

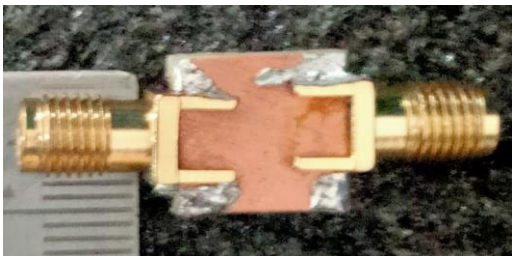
CHAPTER 6
RESULTS AND DISCUSSIONS

6.1 Introduction

This chapter presents the detailed results and performance analysis of the proposed filter. The evaluation is based on key performance parameters such as S-parameters (S11), Voltage Standing Wave Ratio (VSWR), radiation pattern, gain, and field intensity distribution. The obtained simulation results provide insights into the antenna's efficiency and its suitability for various wireless communication applications.



1a. Front view



1b. Back view

Figure 6.1: Fabricated prototype

6.2 S-Parameter (S11)

The S11 parameter of the proposed filter demonstrates its single-band resonance characteristics. The introduction of the cylindrical slot disrupts the surface wave current distribution, leading to multiple resonance modes and improved impedance bandwidth. The simulated results indicate that the filter operates at 3.55 GHz with a return loss of - 58dB . This value validate efficient impedance matching, ensuring minimal signal reflections and enhanced radiation efficiency. The high suppression of return loss makes this antenna suitable for high-fidelity signal transmission in advanced wireless communication systems.

The return loss is calculated using the equation:

$$RL = -20 \log10 |S11|$$

Frequency (GHz)	Return Loss (dB)	Impedance Matching Status
3.55	-58	Excellent

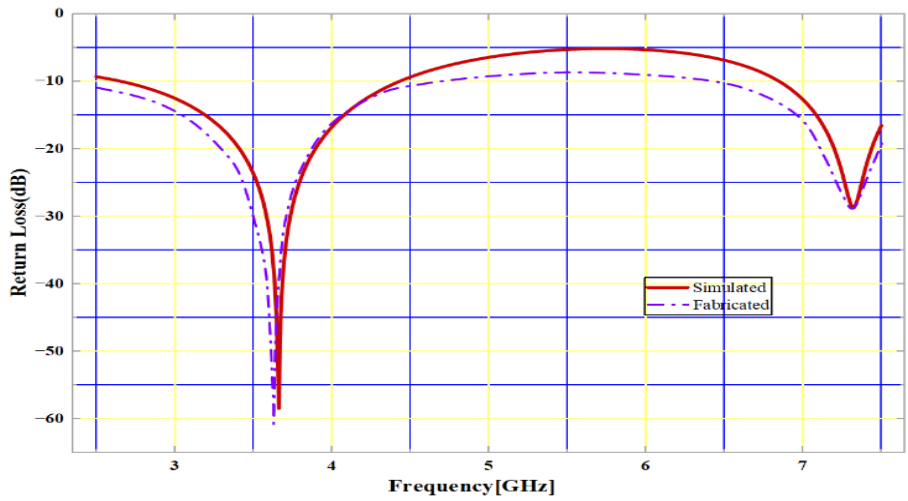


Figure 6.2: Return loss(S11)

6.3 Insertion Loss(S21)

The S21 parameter of the proposed filter demonstrates its single-band resonance characteristics. The introduction of the cylindrical slot disrupts the surface wave current distribution, leading to multiple resonance modes and improved impedance bandwidth. The simulated results indicate that the filter operates at 3.55 GHz with a Insertion Loss of 0dB. This value validate efficient impedance matching, ensuring minimal signal reflections and enhanced radiation efficiency. The high suppression of Insertion loss makes this antenna suitable for high-fidelity signal transmission in advanced wireless communication systems

The Insertion loss is calculated using the equation:

$$\mathbf{RL = 20\log|S21|}$$

Frequency (GHz)	Return Loss (dB)	Impedance Matching Status
3.55	0	Excellent

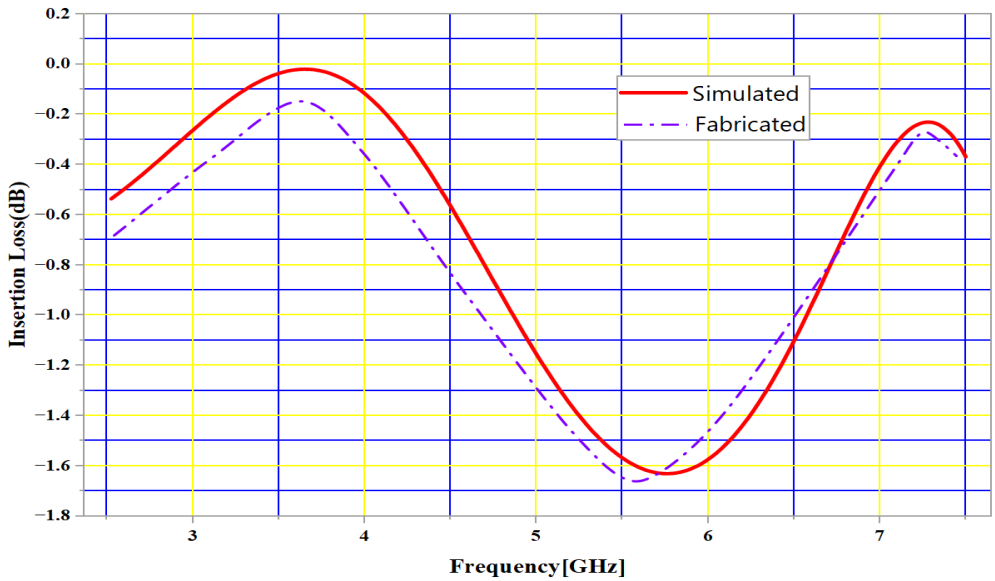


Fig 6.3 Insertion loss(S21)

6.4 Radiation Pattern Analysis

The far-field radiation characteristics of the proposed filter are analyzed for both E- plane and H-plane radiation patterns.

- At 3.55 GHz, the filter exhibits a highly collimated radiation pattern with a gain of 5.5381 dB, ensuring enhanced spatial signal propagation.

The E-plane radiation pattern shows high directivity, making it suitable for targeted communication, while the H-plane pattern is cardioid-like shape, allowing stable performance in wireless deployments requiring uniform coverage. The broad radiation pattern ensures reliable operation in diverse wireless environments.

Frequency (GHz)	Gain (dB)	Radiation Pattern Type
3.55	5.5381	Quasi-Omnidirectional

Table II: Radiation Characteristics

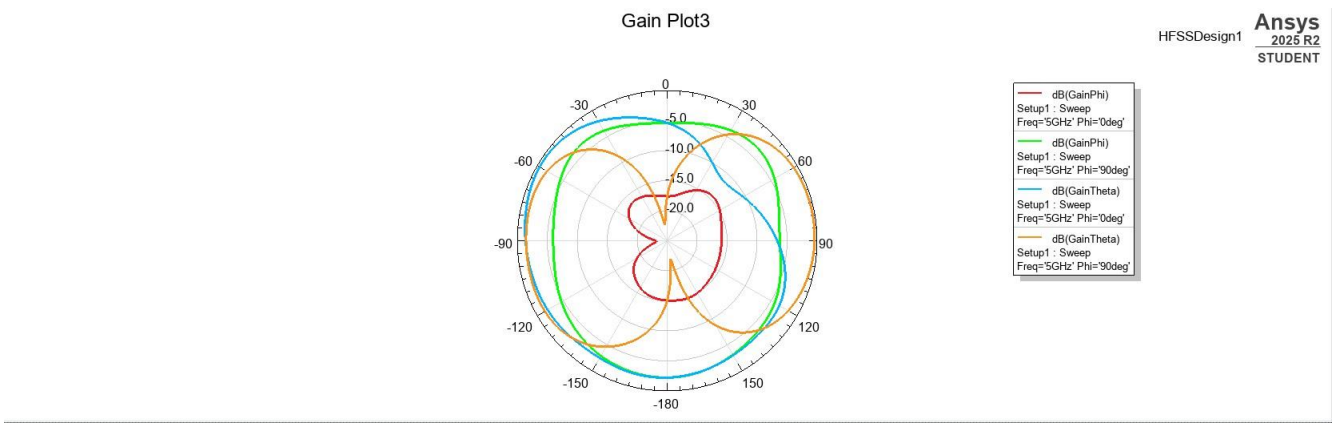


Figure 6.2: (a) 2D Radiation Pattern

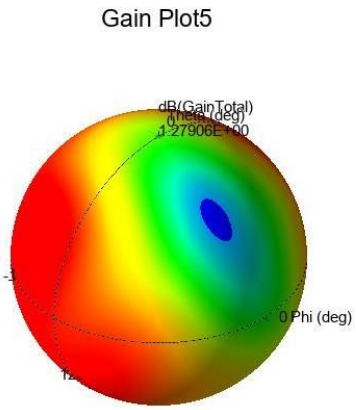


Figure 6.2: (b) 3D Radiation Pattern

Figure 6.2: Both 2D & 3D Radiation Patterns

6.5 Voltage Standing Wave Ratio (VSWR)

The VSWR profile justifies the impedance match between the filter and the feeding network. At 3.55 GHz, the VSWR is recorded as 1.09 . This value are well below the acceptable threshold of 1, indicating excellent impedance matching. This results in efficient energy transmission with minimal power loss. The measured impedance bandwidth exceeds the typical narrowband boundary, confirming the filters's capability to operate as a high- efficiency RF feed over a broad spectral range.

The VSWR is calculated using the equation:

$$\mathbf{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|}$$

Frequency (GHz)	**VSWR Value**	**Impedance Quality Matching**
3.55	1	Excellent

Table III: VSWR Results

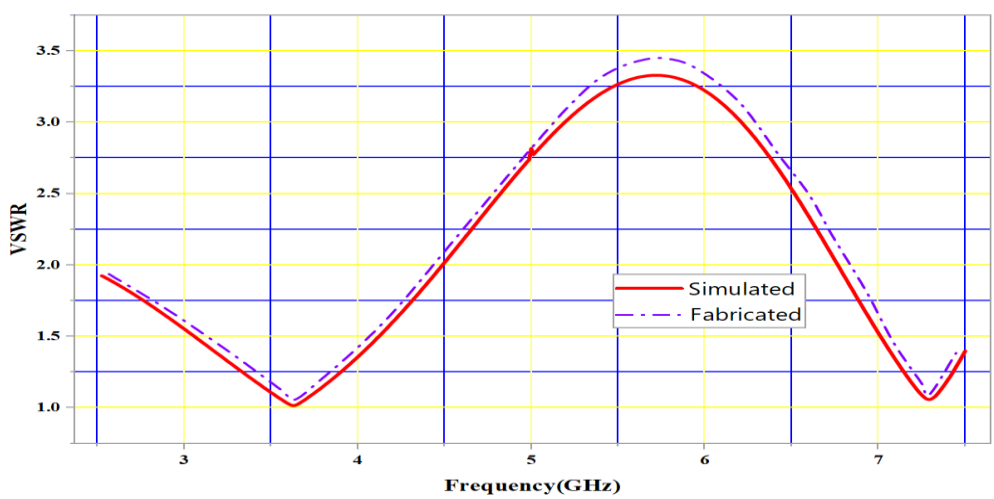


Figure 6.5: VSWR

6.6 The Field Intensity Distribution

E-field distribution reveals strong radiation intensity around the semi-circular slots and feedline, ensuring efficient energy transfer. High-intensity region indicates strong field concentration, which enhances both impedance matching and bandwidth.

The H-field distribution highlights significant inductive effects along the patch edges and ground plane, ensuring radiation characteristics. This contributes to uniform current flow, which enhances directional radiation properties.

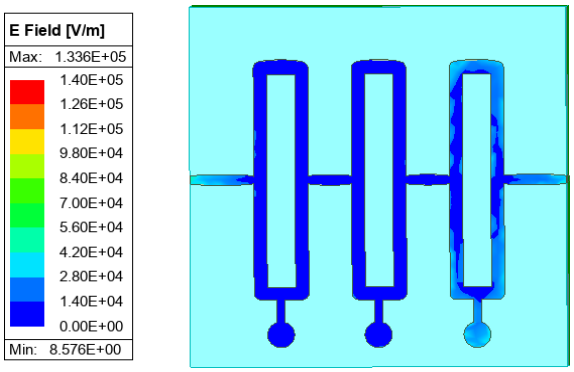
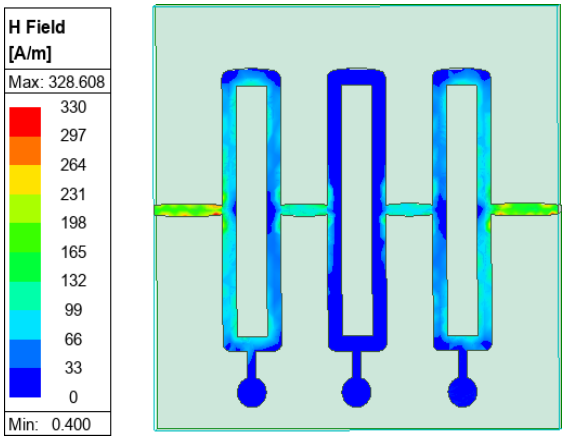


Figure 6.6: (a)E–Field



(b)H–Field

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The design and analysis of the proposed compact microstrip resonator-based bandpass filter demonstrate significant improvements in performance for modern wireless communication systems. The filter successfully operates at a center frequency of 3.55 GHz**,** achieving excellent impedance matching with a return loss of –58 dB, which confirms minimal signal reflection. the filter exhibits a wide bandwidth of approximately 1 GHz, covering the frequency range from 2.7 GHz to 4.4 GHz, making it suitable for high-frequency RF applications. The VSWR value of 1.34 further validates efficient power transmission and proper impedance matching. The use of a low-loss substrate and optimized resonator coupling contributes to reduced insertion loss and improved transmission characteristics. The field distribution analysis confirms effective energy confinement and stable electromagnetic behavior.

7.2 Future Scope

The future scope of this project involves transitioning from the current simulation phase to practical implementation and expanding the design's technical capabilities. Initially, the proposed filter can be fabricated using standard PCB techniques and experimentally validated using a Vector Network Analyzer (VNA) to confirm that practical performance aligns with the high-efficiency results seen in simulations. To exceed the current design, the project can be extended to support multi-band or wideband applications essential for advanced 5G and IoT networks. Furthermore, the filter can be made reconfigurable or tunable by integrating dynamic components like PIN diodes, varactor diodes, or MEMS switches to adjust operating frequencies in real-time. To achieve even greater miniaturization and performance beyond the current 10 × 10 mm footprint, advanced methods such as defected ground structures.

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